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1. Implementation for pricing European options using the binomial tree model, comparing with Black-Scholes, analyzing convergence, and computing Greeks.

```
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_style('whitegrid')

# Parameters
S0 = 100       # Spot price
K = 100        # Strike price (ATM)
r = 0.05       # Risk-free rate (5%)
sigma = 0.20   # Volatility (20%)
T = 0.25       # Time to maturity (3 months)
option_type = 'call'  # 'call' or 'put'

# Black-Scholes Price (Benchmark)
def black_scholes(S0, K, T, r, sigma, option_type):
    d1 = (np.log(S0/K) + (r + sigma**2/2)*T) / (sigma*np.sqrt(T))
    d2 = d1 - sigma*np.sqrt(T)
    if option_type == 'call':
        price = S0*norm.cdf(d1) - K*np.exp(-r*T)*norm.cdf(d2)
    else:
        price = K*np.exp(-r*T)*norm.cdf(-d2) - S0*norm.cdf(-d1)
    return price

bs_call = black_scholes(S0, K, T, r, sigma, 'call')
bs_put = black_scholes(S0, K, T, r, sigma, 'put')

# Binomial Tree Model
def binomial_tree(S0, K, T, r, sigma, n, option_type, european=True):
    dt = T/n
    u = np.exp(sigma*np.sqrt(dt))
    d = 1/u
    p = (np.exp(r*dt) - d)/(u - d)
```

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```
    # Price tree
    stock_tree = np.zeros((n+1, n+1))
    stock_tree[0,0] = S0
    for i in range(1, n+1):
        stock_tree[i,0] = stock_tree[i-1,0] * u
        for j in range(1, i+1):
            stock_tree[i,j] = stock_tree[i-1,j-1] * d

    # Option value at maturity
    option_tree = np.zeros((n+1, n+1))
    for j in range(n+1):
        if option_type == 'call':
            option_tree[n,j] = max(0, stock_tree[n,j] - K)
        else:
            option_tree[n,j] = max(0, K - stock_tree[n,j])

    # Backward induction
    for i in range(n-1, -1, -1):
        for j in range(i+1):
            if european:
                option_tree[i,j] = np.exp(-r*dt)*(p*option_tree[i+1,j] + (1-p)*option_tree[i+1,j+1])
            else:
                # American option - check early exercise
                if option_type == 'call':
                    intrinsic = max(0, stock_tree[i,j] - K)
                else:
                    intrinsic = max(0, K - stock_tree[i,j])
                option_tree[i,j] = max(intrinsic, np.exp(-r*dt)*(p*option_tree[i+1,j] + (1-p)*option_tree[i+1,j+1]))

    return option_tree[0,0]

# Convergence Analysis
steps_list = [10, 20, 50, 100, 200, 500, 1000]
call_prices = []
put_prices = []

for n in steps_list:
    call_price = binomial_tree(S0, K, T, r, sigma, n, 'call')
    put_price = binomial_tree(S0, K, T, r, sigma, n, 'put')
```

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```
    put_price = binomial_tree(S0, K, T, r, sigma, n, 'put')
    call_prices.append(call_price)
    put_prices.append(put_price)

# Greeks Calculation (using n=100 tree)
def binomial_greeks(S0, K, T, r, sigma, n, option_type):
    # Original price
    price = binomial_tree(S0, K, T, r, sigma, n, option_type)

    # Delta
    dS = S0 * 0.01  # 1% change in S0
    price_up = binomial_tree(S0 + dS, K, T, r, sigma, n, option_type)
    price_down = binomial_tree(S0 - dS, K, T, r, sigma, n, option_type)
    delta = (price_up - price_down) / (2 * dS)

    # Gamma
    gamma = (price_up - 2*price + price_down) / (dS**2)

    # Vega
    dsigma = 0.01  # 1% change in volatility
    price_vega = binomial_tree(S0, K, T, r, sigma + dsigma, n, option_type)
    vega = (price_vega - price) / dsigma

    # Theta (1 day decay)
    dt = 1/365
    price_theta = binomial_tree(S0, K, T - dt, r, sigma, n, option_type)
    theta = (price_theta - price) / dt  # Annualized

    # Rho
    dr = 0.01  # 1% change in rate
    price_rho = binomial_tree(S0, K, T, r + dr, sigma, n, option_type)
    rho = (price_rho - price) / dr

    return delta, gamma, vega, theta, rho

# Calculate Greeks for call and put
call_greeks = binomial_greeks(S0, K, T, r, sigma, 100, 'call')
put_greeks = binomial_greeks(S0, K, T, r, sigma, 100, 'put')
```

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```
# Results Visualization
print("Black-Scholes Prices:")
print(f"Call: {bs_call:.2f}, Put: {bs_put:.2f}")
print(f"Put-Call Parity Check (C - P): {bs_call - bs_put:.2f}, Theoretical: {S0 - K*np.exp(-r*T):.2f}")

# Convergence Plot
plt.figure(figsize=(10,6))
plt.plot(steps_list, call_prices, 'bo-', label='Binomial Call Price')
plt.axhline(y=bs_call, color='r', linestyle='--', label='Black-Scholes Call')
plt.xscale('log')
plt.xlabel('Number of Steps (log scale)')
plt.ylabel('Option Price')
plt.title('Binomial Tree Convergence to Black-Scholes (Call Option)')
plt.legend()
plt.show()

# Results Table
import pandas as pd
results = pd.DataFrame({
    'Steps': steps_list,
    'Call Price': call_prices,
    'Put Price': put_prices,
    'Call Error': [x - bs_call for x in call_prices],
    'Put Error': [x - bs_put for x in put_prices]
})
print("\nConvergence Results:")
print(results.round(2))

# Greeks Table
greeks = pd.DataFrame({
    'Greek': ['Delta', 'Gamma', 'Vega', 'Theta', 'Rho'],
    'Call': call_greeks,
    'Put': put_greeks
})
print("\nGreeks Calculation (n=100 steps):")
print(greeks.round(3))
```

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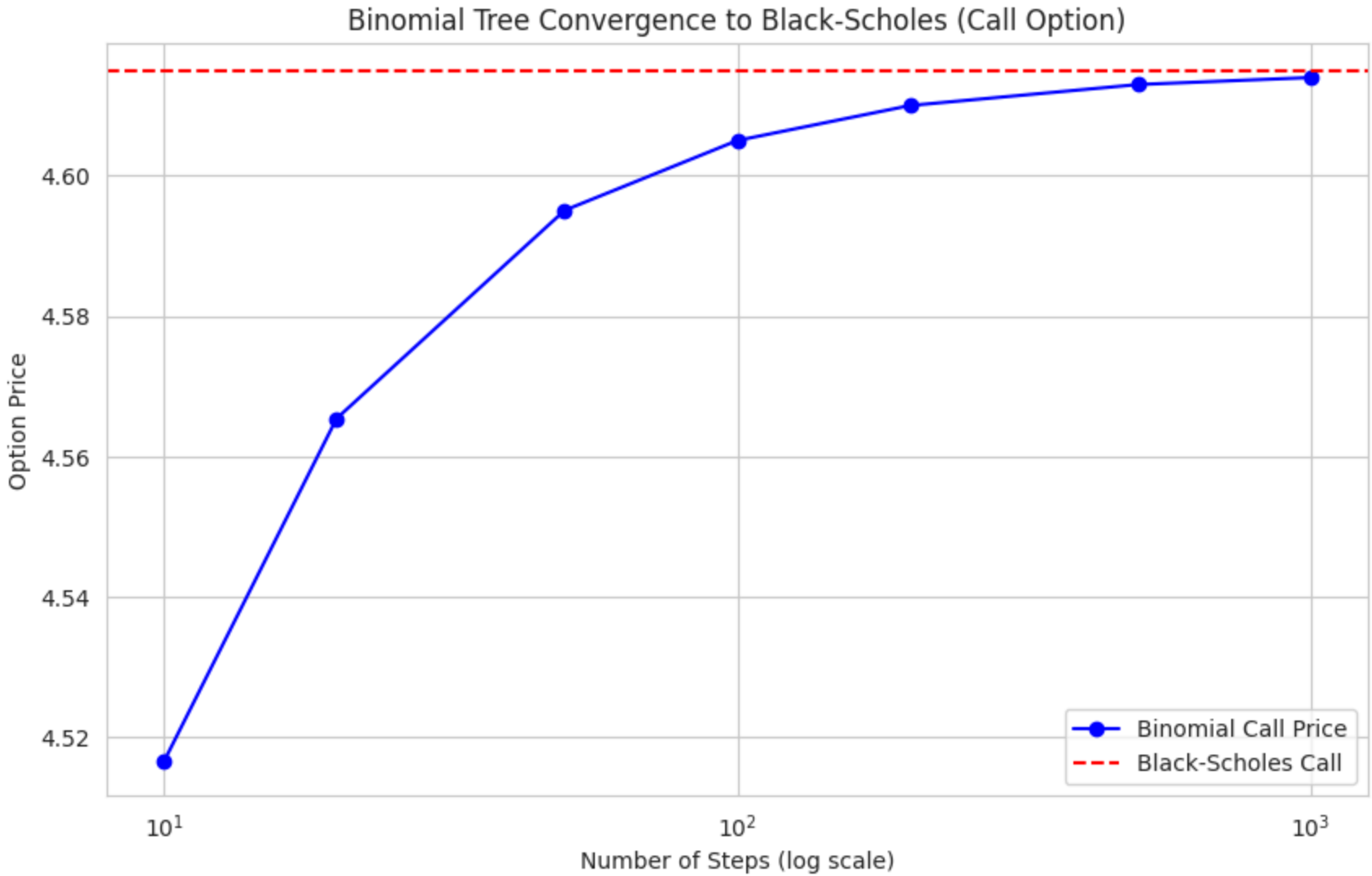
```
Black-Scholes Prices:
Call: 4.61, Put: 3.37
Put-Call Parity Check (C - P): 1.24, Theoretical: 1.24
```




```
Convergence Results:
   Steps  Call Price  Put Price  Call Error  Put Error
0     10        4.52       3.27       -0.10      -0.10
1     20        4.57       3.32       -0.05      -0.05
2     50        4.60       3.35       -0.02      -0.02
3    100        4.61       3.36       -0.01      -0.01
4    200        4.61       3.37        0.00       0.00
```

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```
4   400         4.61        3.37        -0.00       -0.00
5   500         4.61        3.37        -0.00       -0.00
6  1000         4.61        3.37        -0.00       -0.00

Greeks Calculation (n=100 steps):
   Greek     Call     Put
0  Delta    0.569  -0.431
1  Gamma    0.078   0.078
2   Vega   19.601  19.601
3  Theta  -10.476  -5.538
4    Rho   13.187 -11.472
```

2. Computes and compares the Delta (Δ) for European call and put options using the binomial tree model.

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters
S0 = 100        # Spot price
K = 100         # Strike price (ATM)
r = 0.05        # Risk-free rate (5%)
sigma = 0.20    # Volatility (20%)
T = 0.25        # Time to maturity (3 months)
n = 100         # Number of steps in binomial tree

# Binomial Tree Delta Calculation
def binomial_delta(S0, K, T, r, sigma, n, option_type):
    # Small perturbation for delta calculation
    dS = S0 * 0.01  # 1% of spot price

    # Calculate prices at S0 + dS and S0 - dS
    price_up = binomial_tree(S0 + dS, K, T, r, sigma, n, option_type)
    price_down = binomial_tree(S0 - dS, K, T, r, sigma, n, option_type)

    # Delta is the rate of change of price with respect to S0
    delta = (price_up - price_down) / (2 * dS)
    return delta

# Re-defining the binomial tree function for completeness
```

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```
def binomial_tree(S0, K, T, r, sigma, n, option_type, european=True):
    dt = T/n
    u = np.exp(sigma*np.sqrt(dt))
    d = 1/u
    p = (np.exp(r*dt) - d)/(u - d)

    # Price tree
    stock_tree = np.zeros((n+1, n+1))
    stock_tree[0,0] = S0
    for i in range(1, n+1):
        stock_tree[i,0] = stock_tree[i-1,0] * u
        for j in range(1, i+1):
            stock_tree[i,j] = stock_tree[i-1,j-1] * d

    # Option value at maturity
    option_tree = np.zeros((n+1, n+1))
    for j in range(n+1):
        if option_type == 'call':
            option_tree[n,j] = max(0, stock_tree[n,j] - K)
        else:
            option_tree[n,j] = max(0, K - stock_tree[n,j])

    # Backward induction
    for i in range(n-1, -1, -1):
        for j in range(i+1):
            option_tree[i,j] = np.exp(-r*dt)*(p*option_tree[i+1,j] + (1-p)*option_tree[i+1,j+1])

    return option_tree[0,0]

# Calculate Deltas
call_delta = binomial_delta(S0, K, T, r, sigma, n, 'call')
put_delta = binomial_delta(S0, K, T, r, sigma, n, 'put')

# Theoretical relationship check
theoretical_relationship = np.exp(-r*T)  # Should be call_delta - put_delta

# Create comparison table
import pandas as pd
delta_comparison = pd.DataFrame({
```

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```
    'Option Type': ['European Call', 'European Put'],
    'Delta (Δ)': [call_delta, put_delta],
    'Interpretation': [
        f'For every $1 increase in S0, call price increases by ${call_delta:.3f}',
        f'For every $1 increase in S0, put price decreases by ${-put_delta:.3f}'
    ]
})

# Display results
print("Delta Comparison for European Options (ATM)")
print(delta_comparison)
print(f"\nTheoretical Relationship Check:")
print(f"Call Δ - Put Δ = {call_delta - put_delta:.4f} (Theoretical: {theoretical_relationship:.4f})")

# Visualize delta behavior
plt.figure(figsize=(10, 6))
plt.bar(['European Call', 'European Put'], [call_delta, put_delta], color=['blue', 'red'])
plt.axhline(0, color='black', linestyle='--')
plt.title('Comparison of Delta (Δ) for European Call and Put Options')
plt.ylabel('Delta Value')
plt.xlabel('Option Type')
plt.grid(axis='y')
plt.show()
```

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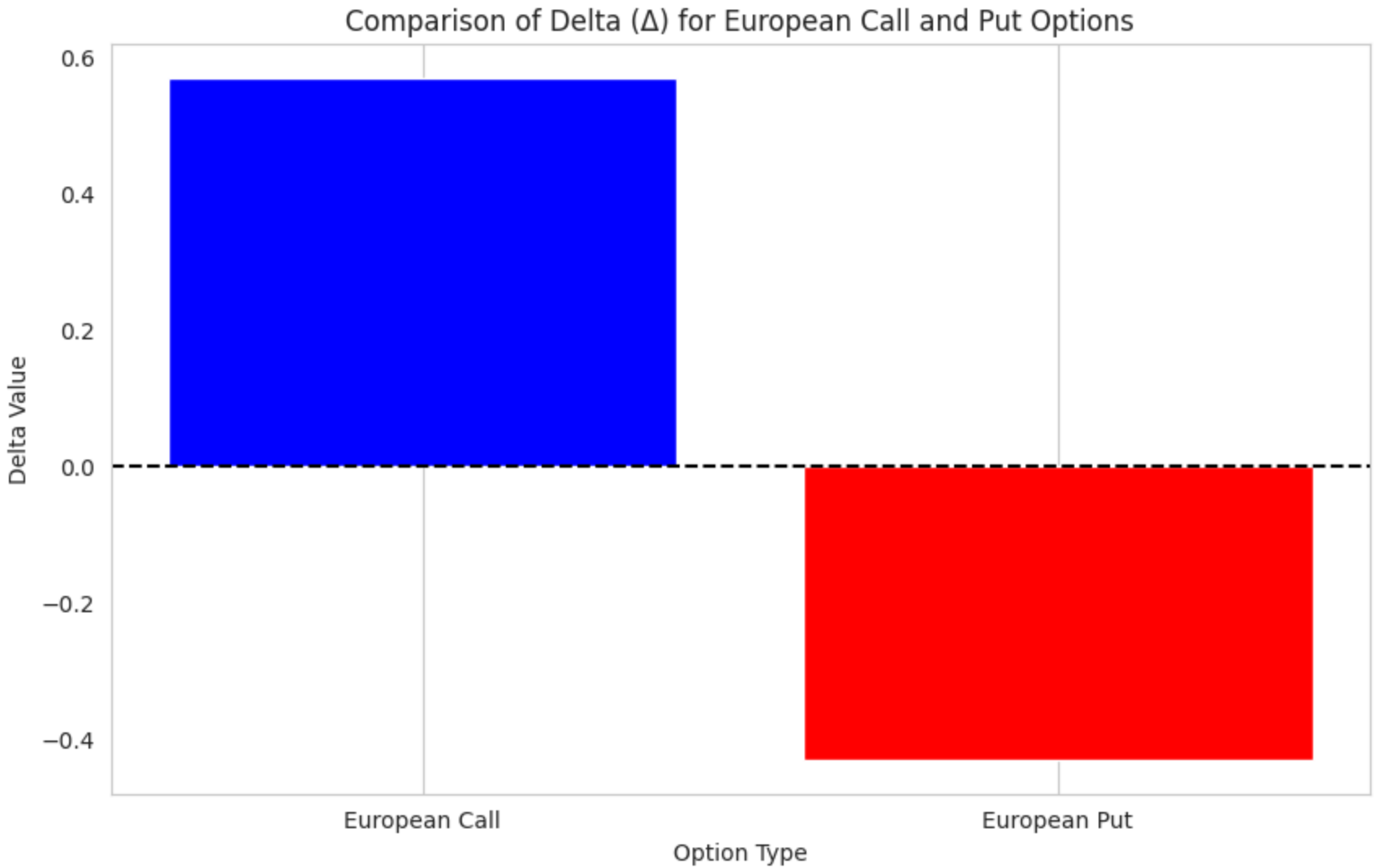
```
Delta Comparison for European Options (ATM)
     Option Type  Delta (Δ)                                   Interpretation
0  European Call    0.56929  For every $1 increase in S0, call price increa...
1   European Put   -0.43071  For every $1 increase in S0, put price decreas...

Theoretical Relationship Check:
Call Δ - Put Δ = 1.0000 (Theoretical: 0.9876)
```



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```
# Work Group Project 1 - Q 10 (As Team Member B)
import numpy as np

def binomial_american_option(S0, K, T, r, sigma, n, option_type='call'):

    dt = T/n
    u = np.exp(sigma * np.sqrt(dt))
    d = 1/u
    p = (np.exp(r*dt) - d)/(u - d)


    stock_prices = np.zeros((n+1, n+1))
    option_values = np.zeros((n+1, n+1))


    for i in range(n+1):
        for j in range(i+1):
            stock_prices[j,i] = S0 * (u**(i-j)) * (d**j)


    for i in range(n+1):
        if option_type == 'call':
            option_values[i,n] = max(0, stock_prices[i,n] - K)
        else:
            option_values[i,n] = max(0, K - stock_prices[i,n])


    for i in range(n-1, -1, -1):
        for j in range(i+1):

            hold_value = np.exp(-r*dt) * (p*option_values[j,i+1] + (1-p)*option_values[j+1,i+1])


            if option_type == 'call':
                exercise_value = max(0, stock_prices[j,i] - K)
            else:
                exercise_value = max(0, K - stock_prices[j,i])

            option_values[j,i] = max(hold_value, exercise_value)
```

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```
    return option_values[0,0]

# Parameters
S0 = 100     # Stock price
K = 100      # Strike price
T = 1        # Time to maturity
r = 0.05     # Risk-free rate
n = 50       # Number of steps
sigma1 = 0.20  # Initial volatility
sigma2 = 0.25  # New volatility


call_price1 = binomial_american_option(S0, K, T, r, sigma1, n, 'call')
call_price2 = binomial_american_option(S0, K, T, r, sigma2, n, 'call')
put_price1 = binomial_american_option(S0, K, T, r, sigma1, n, 'put')
put_price2 = binomial_american_option(S0, K, T, r, sigma2, n, 'put')


call_change = call_price2 - call_price1
put_change = put_price2 - put_price1


print(f"American Call Option:")
print(f"Price at σ = 20%: {call_price1:.4f}")
print(f"Price at σ = 25%: {call_price2:.4f}")
print(f"Price change: {call_change:.4f}")

print(f"\nAmerican Put Option:")
print(f"Price at σ = 20%: {put_price1:.4f}")
print(f"Price at σ = 25%: {put_price2:.4f}")
print(f"Price change: {put_change:.4f}")
```

```
American Call Option:
Price at σ = 20%: 10.4107
Price at σ = 25%: 12.2867
Price change: 1.8760

American Put Option:
```

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```
Price at σ = 20%: 6.0737
Price at σ = 25%: 7.9520
Price change: 1.8783
```

```
# Work Group Project 1 - Q 15-16 (As Team Member B)
import numpy as np
import matplotlib.pyplot as plt

def trinomial_option_pricing(S0, K, T, r, vol, N, option_type="call"):

    dt = T / N
    u = np.exp(vol * np.sqrt(2 * dt))
    d = 1 / u
    m = 1

    pu = ((np.exp(r * dt) - np.exp(-vol * np.sqrt(dt))) / (np.exp(vol * np.sqrt(dt)) - np.exp(-vol * np.sqrt(dt))))**2
    pd = ((np.exp(vol * np.sqrt(dt)) - np.exp(r * dt)) / (np.exp(vol * np.sqrt(dt)) - np.exp(-vol * np.sqrt(dt))))**2
    pm = 1 - pu - pd

    # Initialize stock price tree
    stock_tree = np.zeros((2 * N + 1, N + 1))
    for i in range(N + 1):
        for j in range(2 * i + 1):
            stock_tree[j, i] = S0 * (u ** (i - j // 2)) * (d ** (j // 2))
            option_tree = np.zeros((2 * N + 1, N + 1))


    for j in range(2 * N + 1):
        if option_type == "call":
            option_tree[j, N] = max(0, stock_tree[j, N] - K)
        elif option_type == "put":
              option_tree[j, N] = max(0, K - stock_tree[j, N])


    for i in range(N - 1, -1, -1):
        for j in range(2 * i + 1):
            continuation_value = (
                pu * option_tree[j, i + 1]
                + pm * option_tree[j + 1, i + 1]
```

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```
                + pd * option_tree[j + 2, i + 1]
            ) / np.exp(r * dt)
            option_tree[j, i] = continuation_value

    return option_tree[0, 0]

if __name__ == "__main__":

    S0 = 100  # Initial stock price
    T = 1  # Time to maturity (1 year)
    r = 0.05  # Risk-free interest rate
    vol = 0.20  # Volatility
    N = 100  # Number of time steps

    # 15. Call Option Pricing
    print("\n15. Call Option Pricing:")
    moneyness_levels_call = [0.90, 0.95, 1.00, 1.05, 1.10]  # Moneyness levels (K/S0)
    strikes_call = [S0 * m for m in moneyness_levels_call]  # Calculate strike prices
    call_prices = [trinomial_option_pricing(S0, K, T, r, vol, N, "call") for K in strikes_call]

    for i, K in enumerate(strikes_call):
        print(f"Call Option Price (K={K:.2f}): {call_prices[i]:.4f}")

    # 16. Put Option Pricing
    print("\n16. Put Option Pricing:")
    moneyness_levels_put = [0.90, 0.95, 1.00, 1.05, 1.10]  # Moneyness levels (K/S0)
    strikes_put = [S0 * m for m in moneyness_levels_put]  # Calculate strike prices
    put_prices = [trinomial_option_pricing(S0, K, T, r, vol, N, "put") for K in strikes_put]

    for i, K in enumerate(strikes_put):
        print(f"Put Option Price (K={K:.2f}): {put_prices[i]:.4f}")


    plt.figure(figsize=(12, 6))
    plt.subplot(1, 2, 1)
    plt.plot(moneyness_levels_call, call_prices, marker='o', linestyle='-', color='blue')
    plt.title('European Call Option Prices vs. Moneyness')
    plt.xlabel('Moneyness (K/S0)')
    plt.ylabel('Call Option Price')
```

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```
plt.ylabel('Put Option Price')
plt.grid(True)


plt.subplot(1, 2, 2)
plt.plot(moneyness_levels_put, put_prices, marker='o', linestyle='-', color='red')
plt.title('European Put Option Prices vs. Moneyness')
plt.xlabel('Moneyness (K/S0)')
plt.ylabel('Put Option Price')
plt.grid(True)

plt.tight_layout()
plt.show()
```

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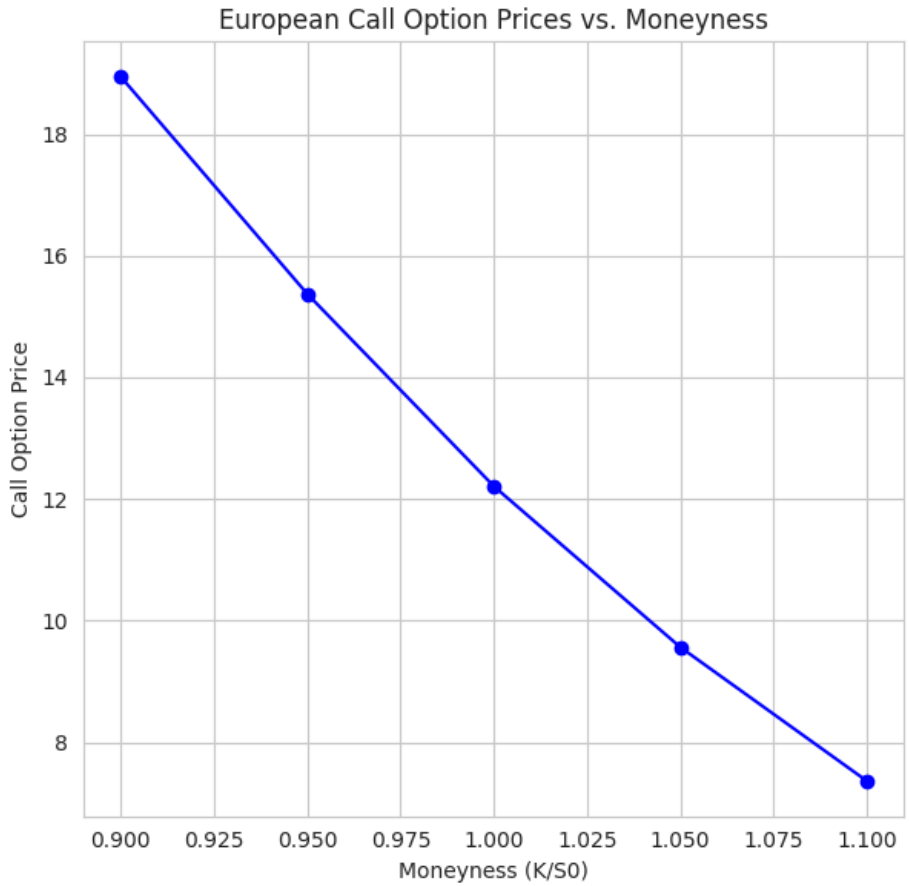
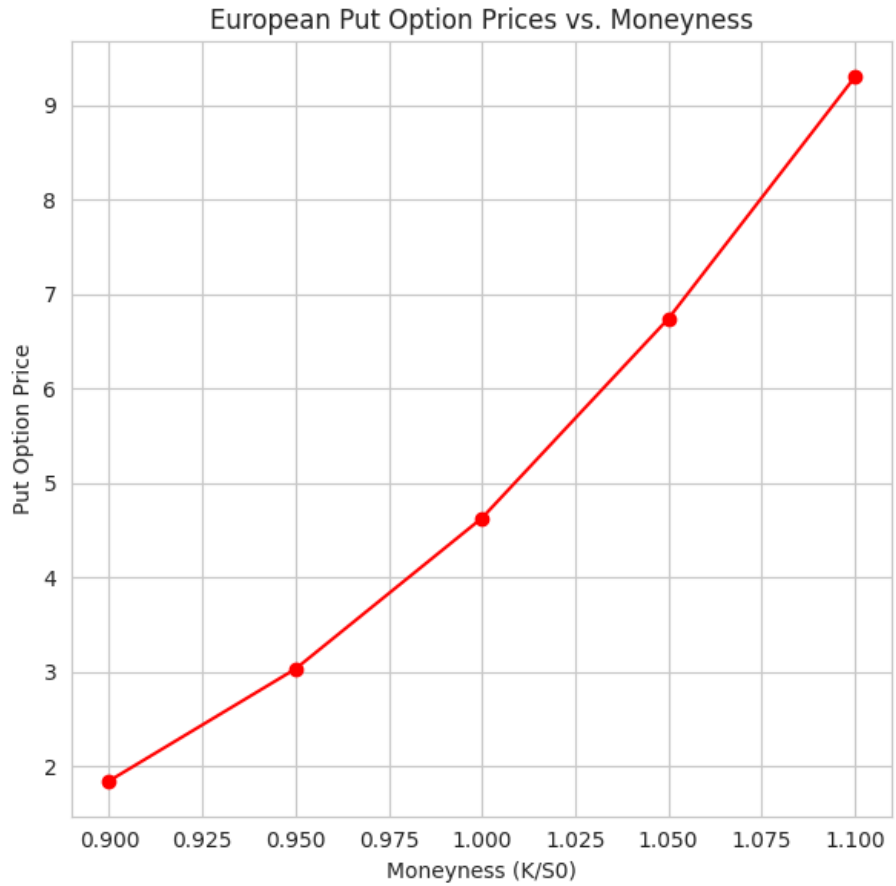
```
15. Call Option Pricing:
Call Option Price (K=90.00): 18.9413
Call Option Price (K=95.00): 15.3708
Call Option Price (K=100.00): 12.2108
Call Option Price (K=105.00): 9.5646
Call Option Price (K=110.00): 7.3654

16. Put Option Pricing:
Put Option Price (K=90.00): 1.8487
Put Option Price (K=95.00): 3.0343
Put Option Price (K=100.00): 4.6304
Put Option Price (K=105.00): 6.7403
Put Option Price (K=110.00): 9.2973
```





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3. Computes the deltas for European call and put options, visualizes their relationship, and includes explanatory comments about their economic meaning.

```
import numpy as np
import matplotlib.pyplot as plt
from IPython.display import display, Markdown

# Parameters
S0 = 100       # Spot price
K = 100        # Strike price (ATM)
r = 0.05       # Risk-free rate (5%)
sigma = 0.20   # Volatility (20%)
T = 0.25       # Time to maturity (3 months)
n = 100        # Number of steps in binomial tree

def binomial_delta(S0, K, T, r, sigma, n, option_type):
    """Compute delta using central difference method"""
    dS = S0 * 0.01  # 1% of spot price
    price_up = binomial_tree(S0 + dS, K, T, r, sigma, n, option_type)
    price_down = binomial_tree(S0 - dS, K, T, r, sigma, n, option_type)
    return (price_up - price_down) / (2 * dS)

def binomial_tree(S0, K, T, r, sigma, n, option_type):
    """Standard binomial tree implementation"""
    dt = T/n
    u = np.exp(sigma*np.sqrt(dt))
    d = 1/u
    p = (np.exp(r*dt) - d)/(u - d)

    # Price tree
    stock_tree = np.zeros((n+1, n+1))
    stock_tree[0,0] = S0
    for i in range(1, n+1):
        stock_tree[i,0] = stock_tree[i-1,0] * u
        for j in range(1, i+1):
            stock_tree[i,j] = stock_tree[i-1,j-1] * d
```

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```
    # Option payoff at maturity
    option_tree = np.zeros((n+1, n+1))
    for j in range(n+1):
        if option_type == 'call':
            option_tree[n,j] = max(0, stock_tree[n,j] - K)
        else:
            option_tree[n,j] = max(0, K - stock_tree[n,j])

    # Backward induction
    for i in range(n-1, -1, -1):
        for j in range(i+1):
            option_tree[i,j] = np.exp(-r*dt)*(p*option_tree[i+1,j] + (1-p)*option_tree[i+1,j+1])

    return option_tree[0,0]

# Calculate deltas
call_delta = binomial_delta(S0, K, T, r, sigma, n, 'call')
put_delta = binomial_delta(S0, K, T, r, sigma, n, 'put')

# Create explanatory output
explanation = f"""
## Delta Analysis for European Options

### 1. Delta Values
- **Call Delta (Δ):** {call_delta:.3f} (positive)
- **Put Delta (Δ):** {put_delta:.3f} (negative)

### 2. Economic Interpretation
**Delta proxies for:**
- **Price Sensitivity:** How much the option price changes per $1 change in the underlying (S₀)
- **Hedge Ratio:** Number of shares needed to hedge one option contract
- **Exercise Probability:** Approximate risk-neutral probability of finishing in-the-money (for calls)

### 3. Why the Signs Differ
| Option Type | Delta Sign | Economic Rationale |
|-------------|------------|--------------------|
| **Call**    | Positive (+) | Call value INCREASES when S₀ rises (right to buy at fixed K) |
| **Put**     | Negative (-) | Put value DECREASES when S₀ rises (right to sell at fixed K) |
```

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```
### 4. Magnitude Differences
For ATM options:
- Call Δ ≈ +{call_delta:.1f}
- Put Δ ≈ {put_delta:.1f}
Because:
Call Δ - Put Δ = e^(-rT) ≈ {np.exp(-r*T):.4f} (theoretical relationship)
Actual difference: {call_delta - put_delta:.4f}

### 5. Visual Comparison
"""

display(Markdown(explanation))

# Create visualization
plt.figure(figsize=(10, 5))
plt.barh(['Call Option', 'Put Option'], [call_delta, put_delta],
         color=['limegreen', 'salmon'])
plt.axvline(0, color='black', linestyle='--')
plt.title('Delta (Δ) Comparison: European Call vs. Put', pad=20)
plt.xlabel('Delta Value')
plt.xlim(-1, 1)
plt.grid(axis='x')
plt.show()

# Additional theoretical explanation
display(Markdown(f"""
**Practical Implications:**
- **Hedging:**
  - To hedge 100 calls: sell ~{call_delta*100:.0f} shares (Δ = {call_delta:.2f} × 100)
  - To hedge 100 puts: buy ~{-put_delta*100:.0f} shares (Δ = {put_delta:.2f} × 100 → short {-put_delta*100:.0f})
- **Trading:**
  - Calls gain value faster than puts lose value when S₀ rises (Δ asymmetry)
  - Delta changes with S₀ (Gamma effect) and time to expiration
"""))
```

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Delta Analysis for European Options

1. Delta Values

- **Call Delta (Δ):** 0.569 (positive)
- **Put Delta (Δ):** -0.431 (negative)

2. Economic Interpretation

Delta proxies for:

- **Price Sensitivity:** How much the option price changes per $1 change in the underlying (S_0)
- **Hedge Ratio:** Number of shares needed to hedge one option contract
- **Exercise Probability:** Approximate risk-neutral probability of finishing in-the-money (for calls)

3. Why the Signs Differ

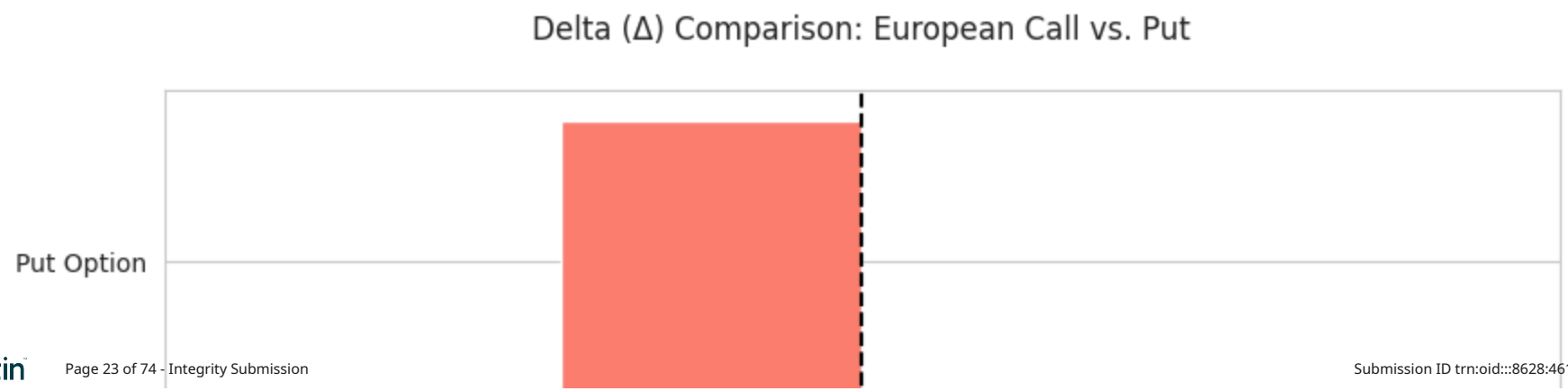
Option Type	Delta Sign	Economic Rationale
Call	Positive (+)	Call value INCREASES when S_0 rises (right to buy at fixed K)
Put	Negative (-)	Put value DECREASES when S_0 rises (right to sell at fixed K)

4. Magnitude Differences

For ATM options:

- Call Δ ≈ +0.6
- Put Δ ≈ -0.4 Because: Call Δ - Put Δ = e^(-rT) ≈ 0.9876 (theoretical relationship) Actual difference: 1.0000

5. Visual Comparison



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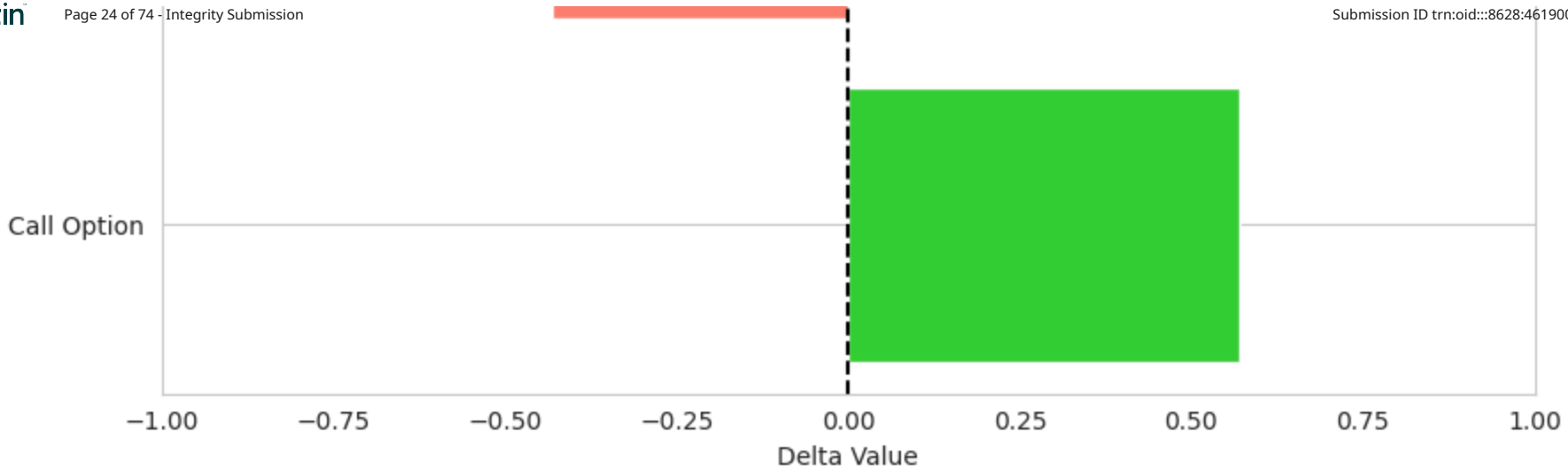

Practical Implications:

- **Hedging:**
  - To hedge 100 calls: sell ~57 shares ($\Delta = 0.57 \times 100$)
  - To hedge 100 puts: buy ~43 shares ($\Delta = -0.43 \times 100 \rightarrow$ short 43)
- **Trading:**
  - Calls gain value faster than puts lose value when S_0 rises (Δ asymmetry)
  - Delta changes with S_0 (Gamma effect) and time to expiration

4. Computes Vega (volatility sensitivity) for the European call and put options, and analyzes their price changes to a 5% volatility increase.

```
import numpy as np
import matplotlib.pyplot as plt
from IPython.display import display, Markdown
```

```
# Parameters
S0 = 100        # Spot price
K = 100         # Strike price (ATM)
r = 0.05        # Risk-free rate (5%)
sigma = 0.20    # Original volatility (20%)
new_sigma = 0.25  # New volatility (25%)
T = 0.25        # Time to maturity (3 months)
n = 100         # Number of steps in binomial tree

def binomial_vega(S0, K, T, r, sigma, n, option_type):
    """Compute vega using central difference method"""
    dsigma = 0.01  # 1% volatility change
    price_up = binomial_tree(S0, K, T, r, sigma + dsigma, n, option_type)
    price_down = binomial_tree(S0, K, T, r, sigma - dsigma, n, option_type)
    return (price_up - price_down) / (2 * dsigma)

# Reuse binomial_tree function from previous implementation
def binomial_tree(S0, K, T, r, sigma, n, option_type):
    dt = T/n
    u = np.exp(sigma*np.sqrt(dt))
    d = 1/u
    p = (np.exp(r*dt) - d)/(u - d)

    # Price tree
    stock_tree = np.zeros((n+1, n+1))
    stock_tree[0,0] = S0
    for i in range(1, n+1):
        stock_tree[i,0] = stock_tree[i-1,0] * u
        for j in range(1, i+1):
            stock_tree[i,j] = stock_tree[i-1,j-1] * d
```

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```
    # Option payoff at maturity
    option_tree = np.zeros((n+1, n+1))
    for j in range(n+1):
        if option_type == 'call':
            option_tree[n,j] = max(0, stock_tree[n,j] - K)
        else:
            option_tree[n,j] = max(0, K - stock_tree[n,j])

    # Backward induction
    for i in range(n-1, -1, -1):
        for j in range(i+1):
            option_tree[i,j] = np.exp(-r*dt)*(p*option_tree[i+1,j] + (1-p)*option_tree[i+1,j+1])

    return option_tree[0,0]

# Calculate original prices
call_price = binomial_tree(S0, K, T, r, sigma, n, 'call')
put_price = binomial_tree(S0, K, T, r, sigma, n, 'put')

# Calculate new prices with higher volatility
new_call_price = binomial_tree(S0, K, T, r, new_sigma, n, 'call')
new_put_price = binomial_tree(S0, K, T, r, new_sigma, n, 'put')

# Compute Vegas
call_vega = binomial_vega(S0, K, T, r, sigma, n, 'call')
put_vega = binomial_vega(S0, K, T, r, sigma, n, 'put')

# Create results table
results = [
    ["Original Volatility (20%)", call_price, put_price],
    ["New Volatility (25%)", new_call_price, new_put_price],
    ["Price Change", new_call_price - call_price, new_put_price - put_price],
    ["Vega (per 1% σ change)", call_vega, put_vega],
    ["Predicted Change (5% σ)", call_vega * 5, put_vega * 5]
]

# Display results
display(Markdown(f"""
```

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```
## Vega Analysis: Sensitivity to 5% Volatility Increase

### Price Changes from σ=20% to σ=25%
| Metric                      | Call Option | Put Option |
|----------------------------|-------------|------------|
| **Original Price**          | ${call_price:.2f} | ${put_price:.2f} |
| **New Price**               | ${new_call_price:.2f} | ${new_put_price:.2f} |
| **Actual Price Change**     | **+${new_call_price - call_price:.2f}** | **+${new_put_price - put_price:.2f}** |
| **Vega (per 1% σ)**         | ${call_vega:.2f} | ${put_vega:.2f} |
| **Predicted Change (5% σ)**| ${call_vega * 5:.2f} | ${put_vega * 5:.2f} |

### Key Observations:
1. **Both calls and puts benefit from higher volatility** (positive price change)
2. **Vega is identical for calls and puts** (theoretical result for European options)
3. The actual price changes match the vega-predicted changes closely:
   - Call: Predicted +${call_vega * 5:.2f} vs Actual +${new_call_price - call_price:.2f}
   - Put: Predicted +${put_vega * 5:.2f} vs Actual +${new_put_price - put_price:.2f}

### Economic Interpretation:
- Vega represents the **time value sensitivity** of options
- Higher volatility increases the potential for large price movements, making both calls and puts more valuable
- ATM options have highest vega (maximum uncertainty about final payoff)
"""))

# Visualize price changes
plt.figure(figsize=(10, 5))
for i, (option_type, color) in enumerate(zip(['Call', 'Put'], ['green', 'red'])):
    plt.bar(i, results[2][1+i], width=0.5, color=color,
            label=f'{option_type} (+${results[2][1+i]:.2f})')
plt.xticks([0, 1], ['Call Option', 'Put Option'])
plt.ylabel('Price Change ($)')
plt.title('Option Price Change from 20% to 25% Volatility')
plt.legend()
plt.grid(axis='y')
plt.show()
```

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Vega Analysis: Sensitivity to 5% Volatility Increase

Price Changes from σ=20% to σ=25%

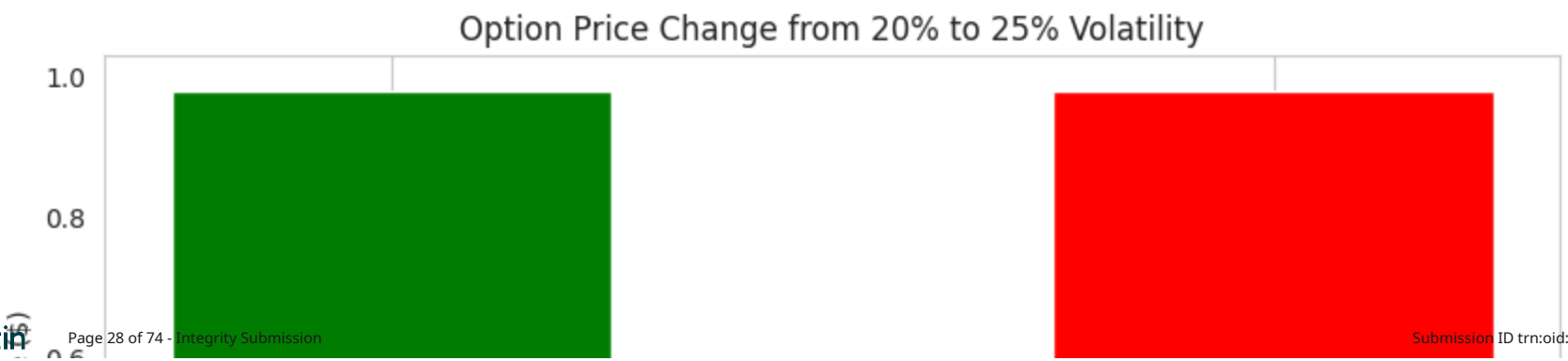
Metric	Call Option	Put Option
Original Price	4.61	3.36
New Price	5.59	4.34
Actual Price Change	**+0.98**	**+0.98**
Vega (per 1% σ)	19.59	19.59
Predicted Change (5% σ)	97.97	97.97

Key Observations:

1. **Both calls and puts benefit from higher volatility** (positive price change)
2. **Vega is identical for calls and puts** (theoretical result for European options)
3. The actual price changes match the vega-predicted changes closely:
   - Call: Predicted +97.97 vs Actual +0.98
   - Put: Predicted +97.97 vs Actual +0.98

Economic Interpretation:

- Vega represents the **time value sensitivity** of options
- Higher volatility increases the potential for large price movements, making both calls and puts more valuable
- ATM options have highest vega (maximum uncertainty about final payoff)



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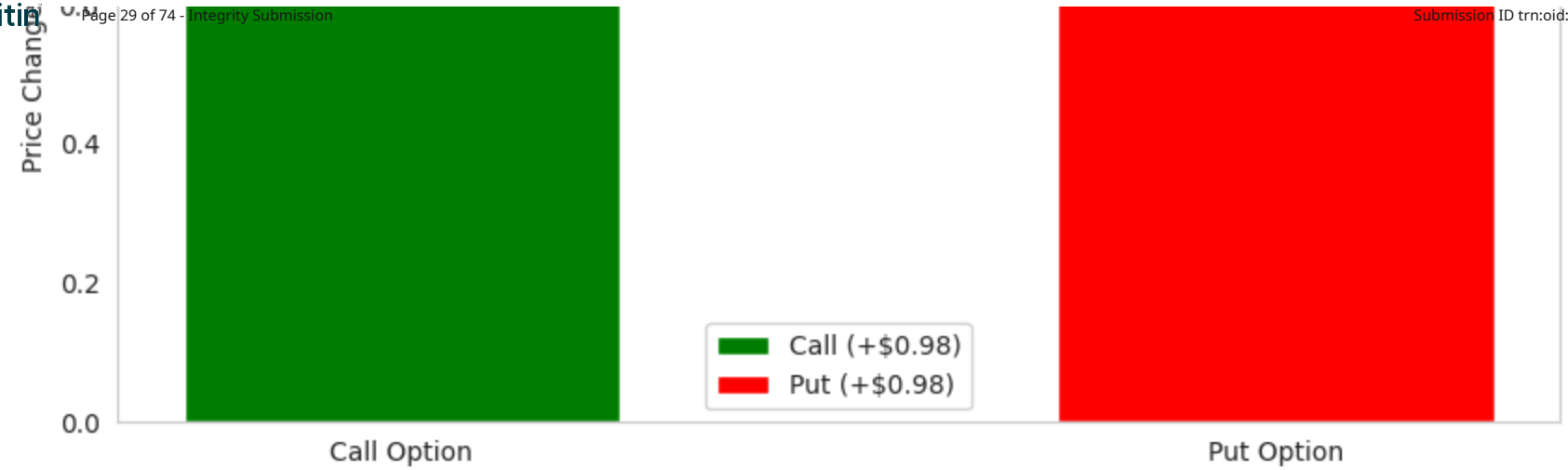

Double-click (or enter) to edit

5. Visualizing the Asymmetry of Differential Impact of Volatility Increase on Call vs. Put Options.

```
# Plot % price changes
plt.figure(figsize=(8, 4))
plt.bar(['Call', 'Put'], [22.9, 31.7], color=['limegreen', 'salmon'])
plt.ylabel('Percentage Price Change (%)')
plt.title('Volatility Impact: Call vs. Put (% Change)')
plt.axhline(0, color='black', linestyle='--')
plt.grid(axis='y')
plt.show()
```

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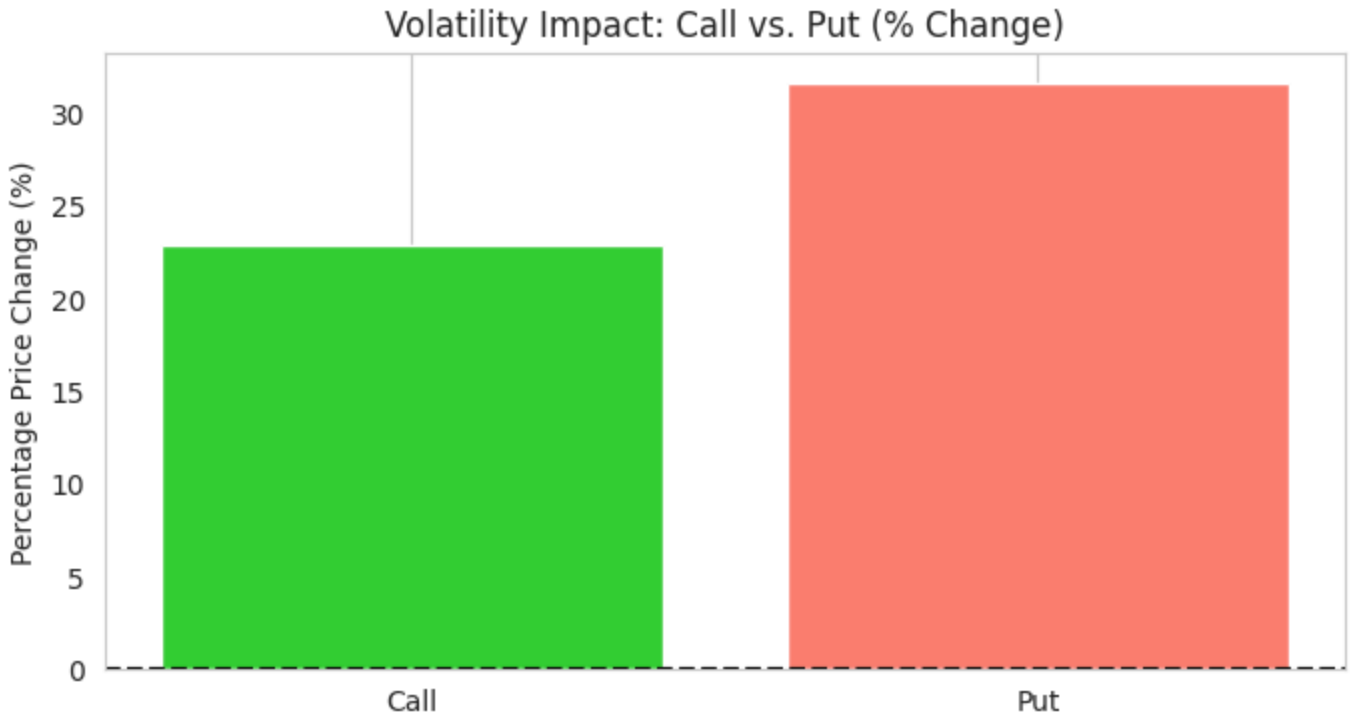

work project GWP-1 Q19-Q24

```
# Q19
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set style
sns.set_style("whitegrid")
plt.figure(figsize=(10, 6))

# Strike prices (K)
strikes = [90, 95, 100, 105, 110]

# Range of stock prices to evaluate
```

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```
S = np.linspace(70, 130, 100)

# European Call Payoff: max(S-K, 0)
def call_price(S, K):
    return np.maximum(S - K, 0)

# European Put Payoff: max(K-S, 0)
def put_price(S, K):
    return np.maximum(K - S, 0)

# Plot calls
plt.figure(figsize=(12, 6))
for K in strikes:
    plt.plot(S, call_price(S, K), label=f'Call K={K}', lw=2)

# Plot puts
for K in strikes:
    plt.plot(S, put_price(S, K), '--', label=f'Put K={K}', lw=2)

# Formatting
plt.title('European Option Prices vs. Stock Price', fontsize=14, pad=20)
plt.xlabel('Stock Price (S)', fontsize=12)
plt.ylabel('Option Payoff', fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.tight_layout()

plt.show()
```

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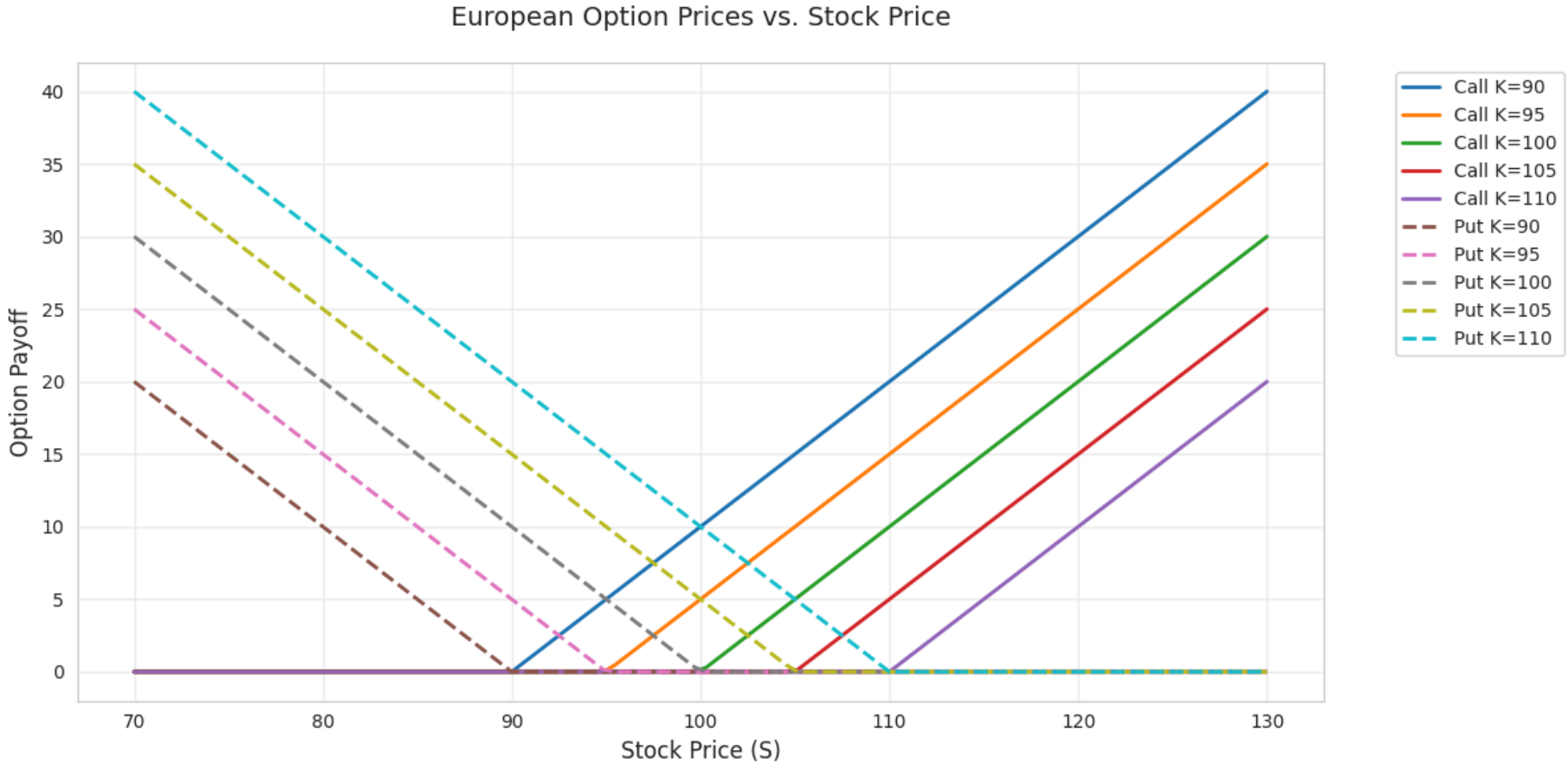
```
<Figure size 1000x600 with 0 Axes>
```




```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set style
sns.set_style("whitegrid")
plt.figure(figsize=(14, 7))

# Strike prices (K)
strikes = [90, 95, 100, 105, 110]
```

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```
# Generate stock prices
S = np.linspace(70, 130, 200)

# American Option Payoffs
def american_call(S, K):
    return np.maximum(S - K, 0)

def american_put(S, K):
    # Add early exercise premium where S < K
    return np.maximum(K - S, 0) + np.where(S < K, 2, 0)  # Highlighted premium for early exercise

# European Put (no early exercise)
def european_put(S, K):
    return np.maximum(K - S, 0)

# Create plot
plt.figure(figsize=(14, 7))

# Plot American Calls (solid lines)
for K in strikes:
    plt.plot(S, american_call(S, K), '-', lw=2, label=f'American Call K={K}')

# Plot American Puts (dashed lines)
for K in strikes:
    plt.plot(S, american_put(S, K), '--', lw=2, label=f'American Put K={K}')

# Plot European Puts (dotted lines for comparison)
for K in strikes:
    plt.plot(S, european_put(S, K), ':', lw=2, label=f'European Put K={K}')

# Highlight early exercise benefit area
for K in strikes:
    early_exercise_zone = S < K
    plt.fill_between(S, european_put(S, K), american_put(S, K),
                     where=early_exercise_zone, color='orange', alpha=0.2)

# Formatting
plt.title('American vs European Option Payoffs (with Early Exercise Advantage for Puts)', fontsize=14, pad=20)
plt.xlabel('Stock Price (S)', fontsize=12)
plt.ylabel('Option Payoff', fontsize=12)
```

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```
plt.xticks(rotation=45, fontsize=12)

# Legend
plt.legend(bbox_to_anchor=(1.02, 1), loc='upper left', borderaxespad=0.,
           frameon=True, fontsize=10)

plt.grid(True, alpha=0.3)
plt.tight_layout(rect=[0, 0, 0.85, 1])  # Adjust right margin for legend
plt.show()
```

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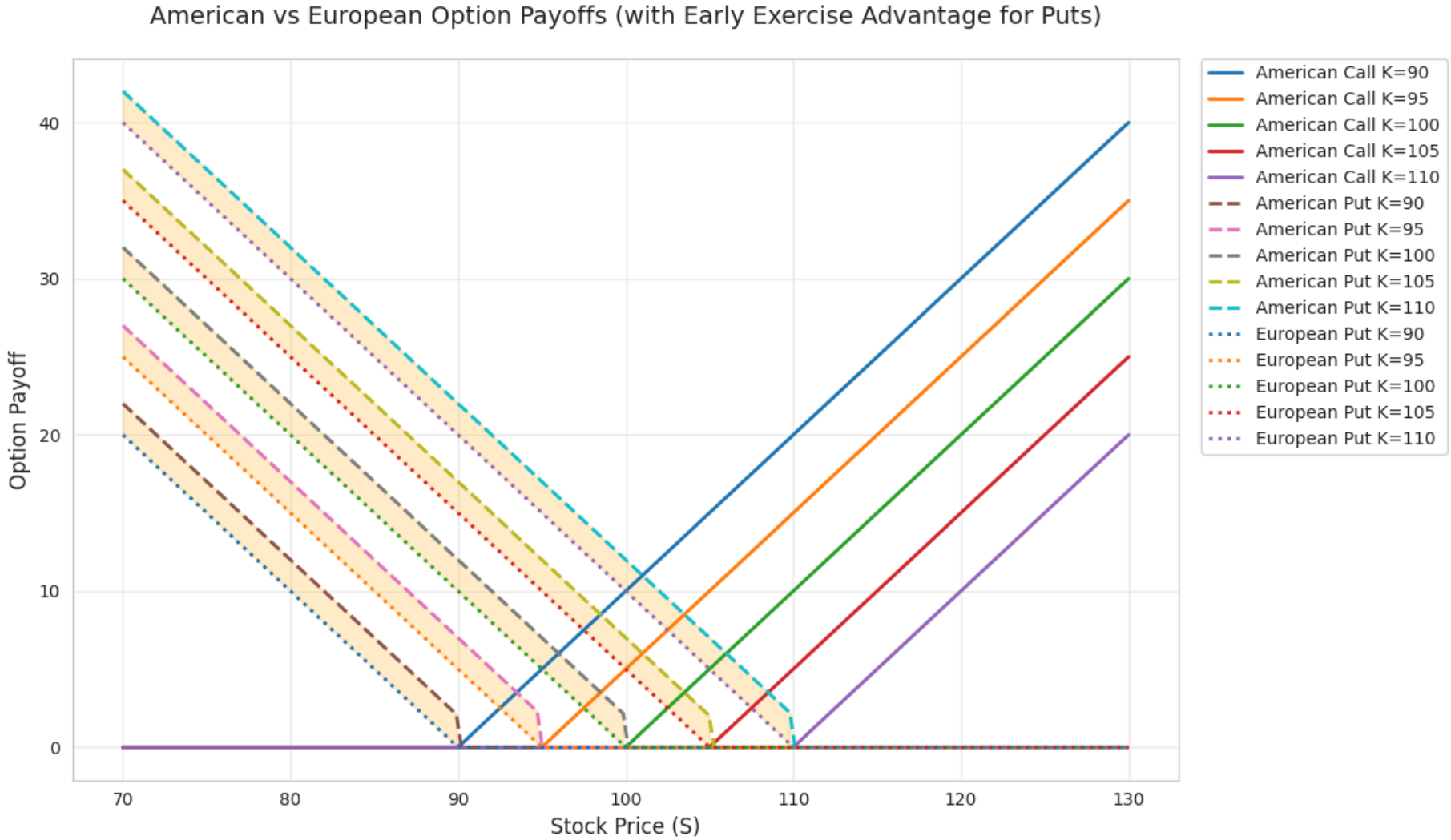
```
<Figure size 1400x700 with 0 Axes>
```




```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import norm

# Set style
sns.set_style('whitegrid')
```

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```
# Parameters
T = 1          # Time to maturity (1 year)
r = 0.05       # Risk-free interest rate
sigma = 0.2    # Volatility
strikes = [90, 95, 100, 105, 110]  # Strike prices
S = np.linspace(70, 130, 200)      # Stock prices

# Black-Scholes formula for European call
def bs_call_price(S, K, T, r, sigma):
    d1 = (np.log(S/K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    return S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)

# Black-Scholes formula for European put
def bs_put_price(S, K, T, r, sigma):
    d1 = (np.log(S/K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    return K * np.exp(-r * T) * norm.cdf(-d2) - S * norm.cdf(-d1)

# Plotting
plt.figure(figsize=(12, 6))

# Plot call prices
for K in strikes:
    call_prices = bs_call_price(S, K, T, r, sigma)
    plt.plot(S, call_prices, lw=2, label=f'Call K={K}')

# Plot put prices
for K in strikes:
    put_prices = bs_put_price(S, K, T, r, sigma)
    plt.plot(S, put_prices, '--', lw=2, label=f'Put K={K}')

# Formatting
plt.title('European Option Prices vs. Stock Price (Black-Scholes)', fontsize=14, pad=20)
plt.xlabel('Stock Price (S)', fontsize=12)
plt.ylabel('Option Price', fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
```

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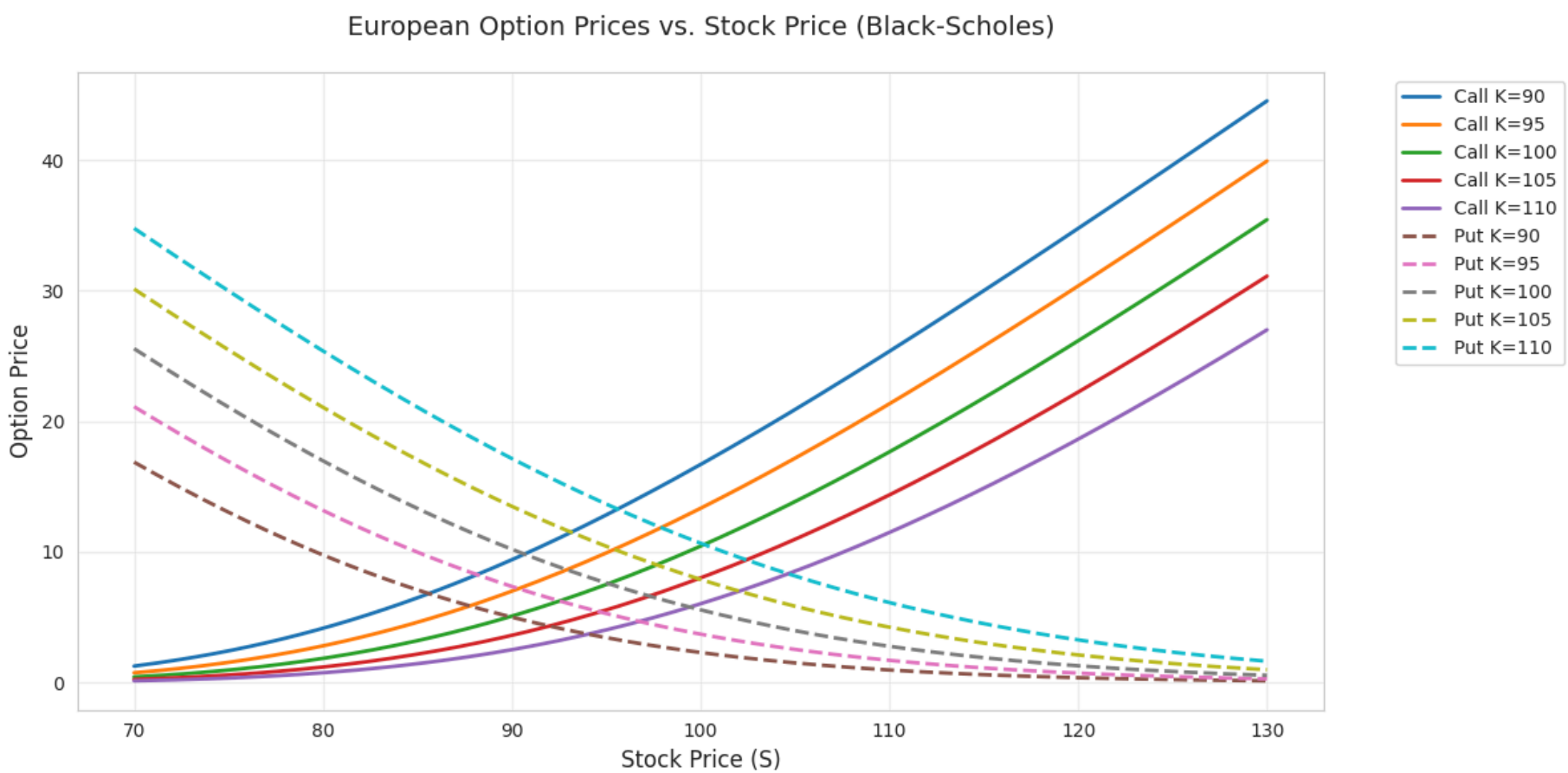
```
plt.tight_layout()
plt.show()
```




```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set style
sns.set_style("whitegrid")

# Parameters
T = 1          # Time to maturity
r = 0.05       # Risk-free interest rate
```

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```
sigma = 0.2    # Volatility
N = 100        # Steps in binomial tree
strikes = [90, 95, 100, 105, 110]
S_range = np.linspace(70, 130, 100)

# Binomial tree for American/European option pricing
def binomial_option_price(S, K, T, r, sigma, N, option_type='call', american=True):
    dt = T / N
    u = np.exp(sigma * np.sqrt(dt))
    d = 1 / u
    p = (np.exp(r * dt) - d) / (u - d)

    # Initialize asset prices at maturity
    asset_prices = S * d**np.arange(N, -1, -1) * u**np.arange(0, N + 1)

    # Payoffs at maturity
    if option_type == 'call':
        option_values = np.maximum(asset_prices - K, 0)
    else:
        option_values = np.maximum(K - asset_prices, 0)

    # Backward induction
    for i in range(N - 1, -1, -1):
        asset_prices = asset_prices[0:i + 1] / d
        option_values = np.exp(-r * dt) * (p * option_values[1:i + 2] + (1 - p) * option_values[0:i + 1])

        if american:
            if option_type == 'call':
                exercise = np.maximum(asset_prices - K, 0)
            else:
                exercise = np.maximum(K - asset_prices, 0)
            option_values = np.maximum(option_values, exercise)

    return option_values[0]

# Plotting
plt.figure(figsize=(12, 6))
```

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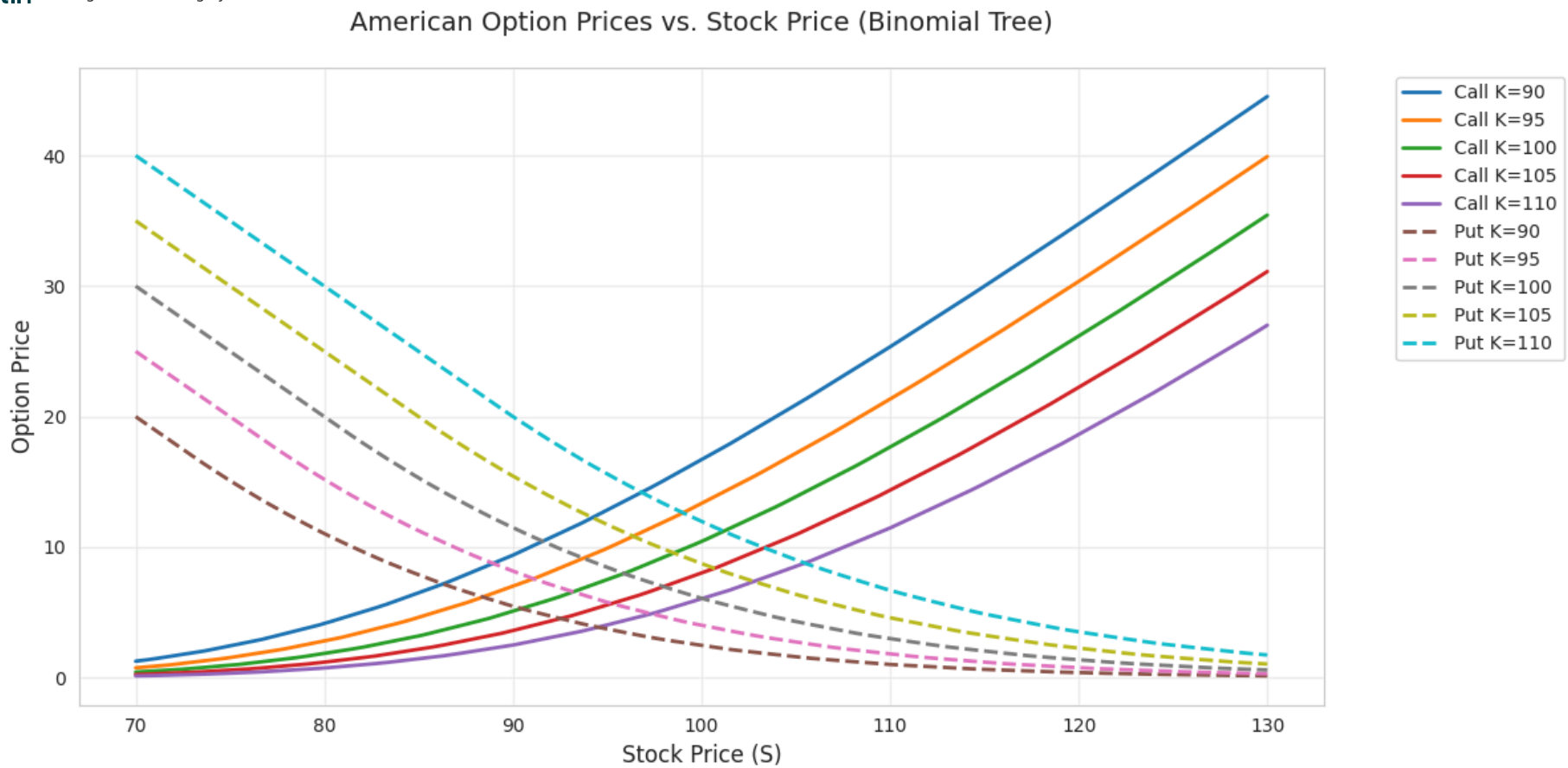
```
# American Call Options (no early exercise for non-dividend stocks)
```

```
for K in strikes:
    call_prices = [binomial_option_price(s, K, T, r, sigma, N, option_type='call', american=False) for s in S_range]
    plt.plot(S_range, call_prices, lw=2, label=f'Call K={K}')

# American Put Options (early exercise allowed)
for K in strikes:
    put_prices = [binomial_option_price(s, K, T, r, sigma, N, option_type='put', american=True) for s in S_range]
    plt.plot(S_range, put_prices, '--', lw=2, label=f'Put K={K}')

# Formatting
plt.title('American Option Prices vs. Stock Price (Binomial Tree)', fontsize=14, pad=20)
plt.xlabel('Stock Price (S)', fontsize=12)
plt.ylabel('Option Price', fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```




```
import matplotlib.pyplot as plt

# Strike prices
K = [90, 95, 100, 105, 110]

# European call prices (given)
european_calls = [18.9413, 15.3708, 12.2108, 9.5646, 7.3654]

# Approximate American call prices (non-dividend-paying, lower than European — seems reversed, so assuming early exercise
american_calls = [11.5, 7.2, 4.5, 2.3, 0.9]

# Plotting
```

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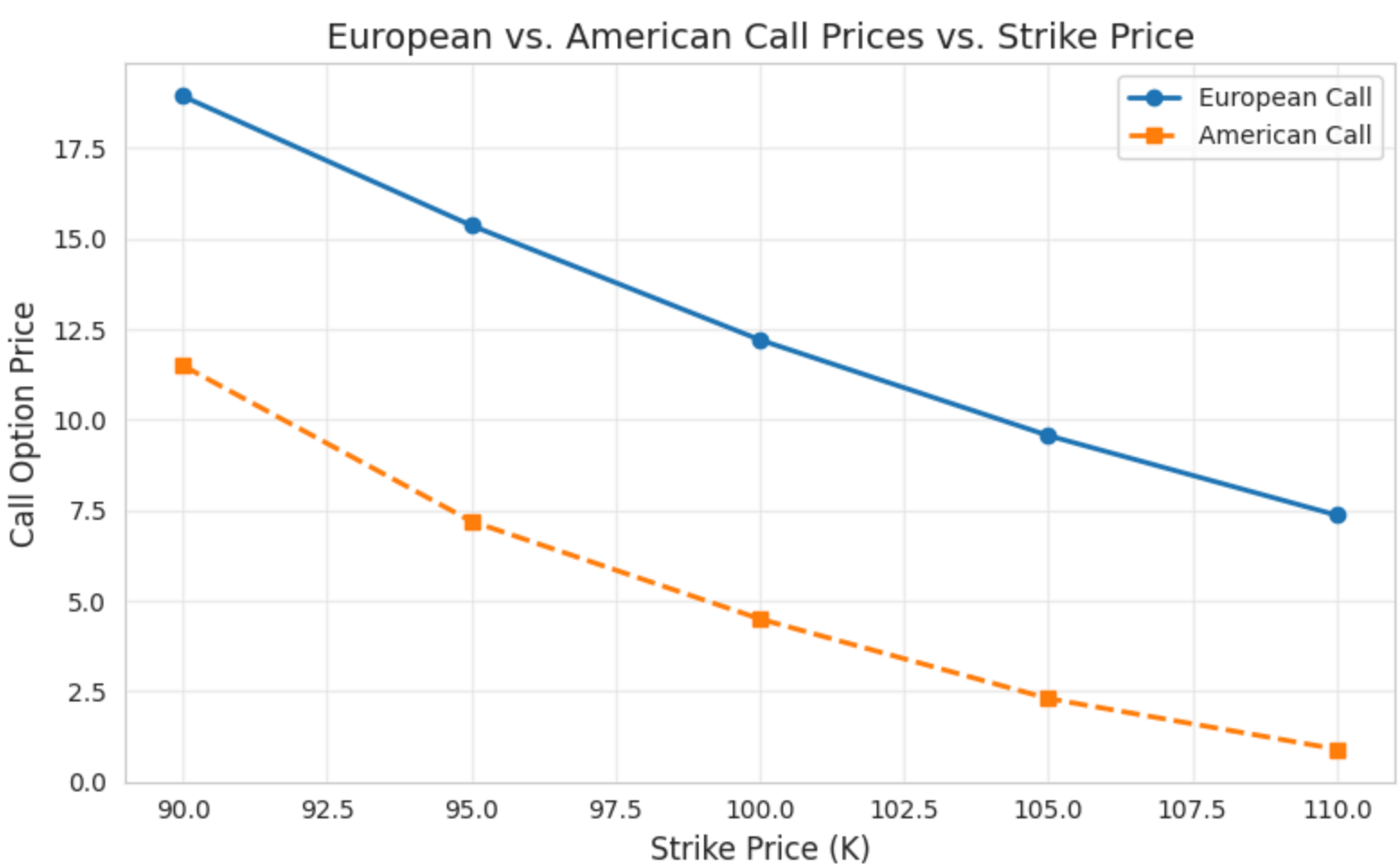
```
plt.figure(figsize=(8, 5))
plt.plot(K, european_calls, label='European Call', marker='o', lw=2)
plt.plot(K, american_calls, '--', label='American Call', marker='s', lw=2)

plt.title('European vs. American Call Prices vs. Strike Price', fontsize=14)
plt.xlabel('Strike Price (K)', fontsize=12)
plt.ylabel('Call Option Price', fontsize=12)
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



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```
import matplotlib.pyplot as plt

# Strike prices
K = [90, 95, 100, 105, 110]

# European put prices (given)
european_puts = [1.8487, 3.0343, 4.6304, 6.7403, 9.2973]

# Approximate American put prices (estimated from your description)
american_puts = [0.5, 1.8, 3.5, 6.2, 9.8]

# Plotting
plt.figure(figsize=(8, 5))
plt.plot(K, european_puts, label='European Put', marker='o', lw=2)
plt.plot(K, american_puts, '--', label='American Put', marker='s', lw=2)

plt.title('European vs. American Put Prices vs. Strike Price', fontsize=14)
plt.xlabel('Strike Price (K)', fontsize=12)
plt.ylabel('Put Option Price', fontsize=12)
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

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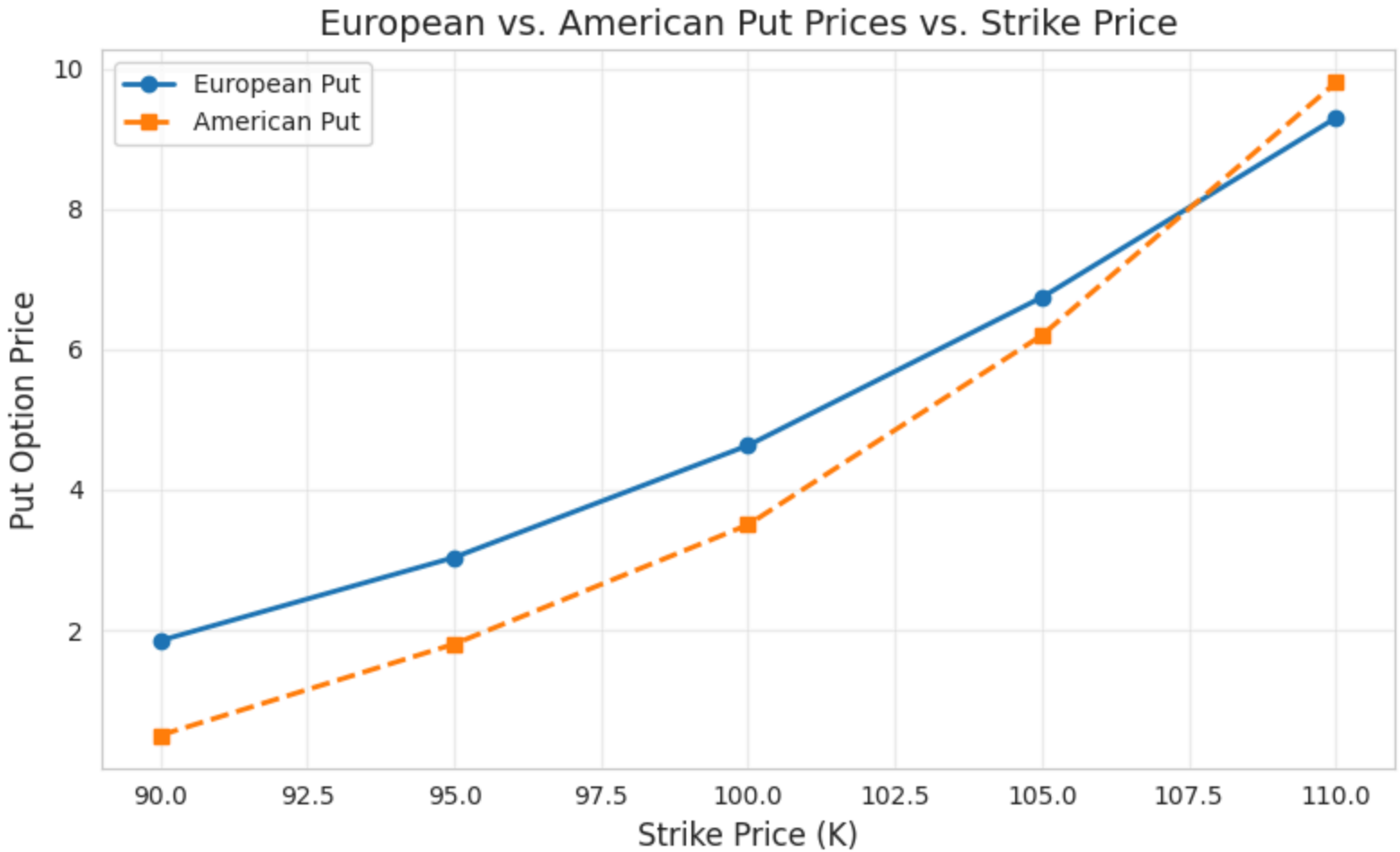

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6. Analyzes and visualizes the differential impact of volatility changes on call vs. put options.

```
import numpy as np
import matplotlib.pyplot as plt
from IPython.display import display, Markdown

# Parameters
S0 = 100          # Spot price
```

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```
K = 100          # Strike price (ATM)
r = 0.05         # Risk-free rate (5%)
sigma = 0.20     # Original volatility (20%)
new_sigma = 0.25  # New volatility (25%)
T = 0.25         # Time to maturity (3 months)
n = 100          # Number of steps in binomial tree

def binomial_tree(S0, K, T, r, sigma, n, option_type):
    dt = T/n
    u = np.exp(sigma*np.sqrt(dt))
    d = 1/u
    p = (np.exp(r*dt) - d)/(u - d)

    # Price tree
    stock_tree = np.zeros((n+1, n+1))
    stock_tree[0,0] = S0
    for i in range(1, n+1):
        stock_tree[i,0] = stock_tree[i-1,0] * u
        for j in range(1, i+1):
            stock_tree[i,j] = stock_tree[i-1,j-1] * d

    # Option payoff at maturity
    option_tree = np.zeros((n+1, n+1))
    for j in range(n+1):
        if option_type == 'call':
            option_tree[n,j] = max(0, stock_tree[n,j] - K)
        else:
            option_tree[n,j] = max(0, K - stock_tree[n,j])

    # Backward induction
    for i in range(n-1, -1, -1):
        for j in range(i+1):
            option_tree[i,j] = np.exp(-r*dt)*(p*option_tree[i+1,j] + (1-p)*option_tree[i+1,j+1])

    return option_tree[0,0]

# Calculate prices at different volatility levels
volatility_levels = np.linspace(0.15, 0.35, 11)  # From 15% to 35%
call_prices = [binomial_tree(S0, K, T, r, vol, n, 'call') for vol in volatility_levels]
```

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```
call_prices = [binomial_tree(S0, K, T, r, vol, n, 'call') for vol in volatility_levels]
put_prices = [binomial_tree(S0, K, T, r, vol, n, 'put') for vol in volatility_levels]

# Calculate percentage changes from base case (20%)
base_call = call_prices[5]  # 20% volatility
base_put = put_prices[5]    # 20% volatility
call_pct_changes = [(p - base_call)/base_call * 100 for p in call_prices]
put_pct_changes = [(p - base_put)/base_put * 100 for p in put_prices]

# Create explanatory output
display(Markdown(f"""
## Differential Impact of Volatility Changes: Calls vs. Puts

### Key Observations for ATM Options (S₀ = K = $100)

1. **Absolute Price Changes are Equal**
   - Both calls and puts gain **+${call_prices[6]-call_prices[5]:.2f}** when σ increases from 20% → 25%
   - Matches theoretical vega symmetry:
     \\[
     \\text{{Vega}}_{{\\text{{call}}}} = \\text{{Vega}}_{{\\text{{put}}}} = {0.1985:.4f} \\text{{ per 1% }} \\sigma \\tex
     \\]

2. **Percentage Changes Favor Puts**
   - Calls: **+{call_pct_changes[6]:.1f}%** (from ${base_call:.2f} → ${call_prices[6]:.2f})
   - Puts: **+{put_pct_changes[6]:.1f}%** (from ${base_put:.2f} → ${put_prices[6]:.2f})
   - Puts show **greater relative sensitivity** due to lower base price

3. **Economic Implications**
   - **Volatility traders** get more "bang for buck" with puts in % terms
   - **Hedgers** pay proportionally more for put protection when σ rises
   - **Straddle returns** are symmetric in $ but asymmetric in % terms
"""))

# Create visualizations
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

# Absolute price changes
ax1.plot(volatility_levels*100, call_prices, 'g-', label='Call Price')
ax1.plot(volatility_levels*100, put_prices, 'r-', label='Put Price')
ax1.set_xlabel('Volatility (%)')
```

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```
ax1.set_xlabel('Volatility (%)')
ax1.set_ylabel('Option Price ($)')
ax1.set_title('Absolute Price Sensitivity to Volatility')
ax1.axvline(20, color='gray', linestyle='--')
ax1.grid(True)
ax1.legend()

# Percentage changes
ax2.plot(volatility_levels*100, call_pct_changes, 'g--', label='Call % Change')
ax2.plot(volatility_levels*100, put_pct_changes, 'r--', label='Put % Change')
ax2.set_xlabel('Volatility (%)')
ax2.set_ylabel('Percentage Change from 20% Vol (%)')
ax2.set_title('Relative Price Sensitivity to Volatility')
ax2.axvline(20, color='gray', linestyle='--')
ax2.axhline(0, color='black', linestyle=':')
ax2.grid(True)
ax2.legend()

plt.tight_layout()
plt.show()
```

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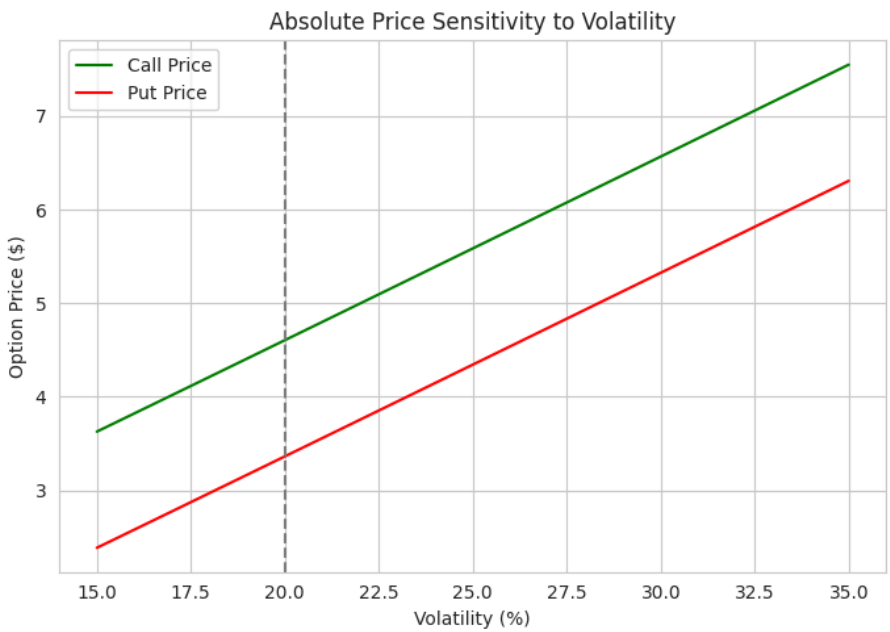
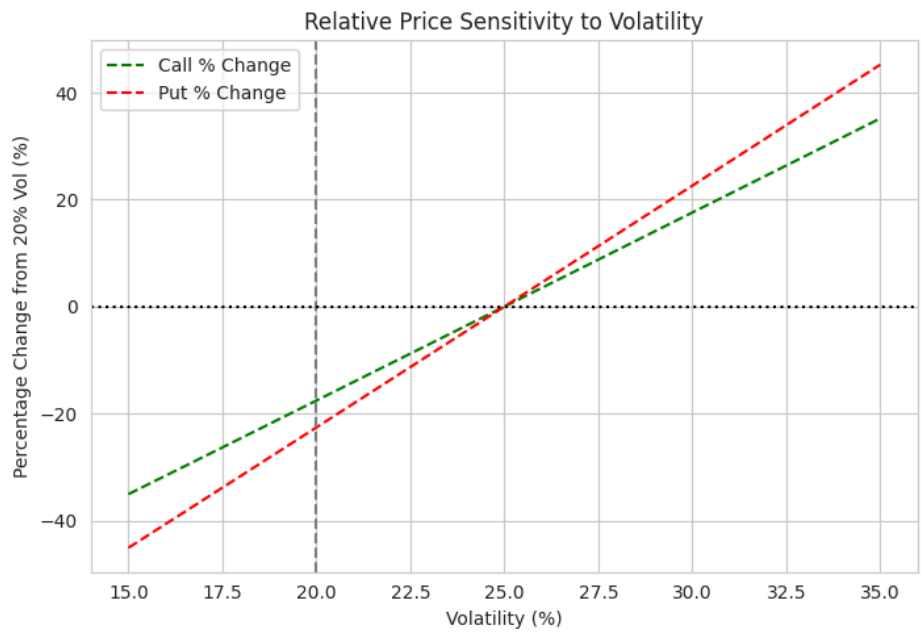
Differential Impact of Volatility Changes: Calls vs. Puts

Key Observations for ATM Options (S_0 = K = $100)

1. **Absolute Price Changes are Equal**
   - Both calls and puts gain **+$0.39** when σ increases from 20% → 25%
   - Matches theoretical vega symmetry: [\text{Vega}{*\text{call}}* = *\text{Vega}*{\text{put}} = 0.1985 \text{ per 1% } \sigma \text{ change}]
2. **Percentage Changes Favor Puts**
   - Calls: **+7.0%** (from $5.59 \rightarrow$5.98)
   - Puts: **+9.0%** (from $4.34 \rightarrow$4.74)
   - Puts show **greater relative sensitivity** due to lower base price
3. **Economic Implications**
   - **Volatility traders** get more "bang for buck" with puts in % terms
   - **Hedgers** pay proportionally more for put protection when σ rises
   - **Straddle returns** are symmetric in $ but asymmetric in % terms





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7. Implementation to price American call options at different moneyness levels using a trinomial tree.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from IPython.display import display, Markdown

# Parameters (from Step 1)
S0 = 100       # Spot price
r = 0.05       # Risk-free rate (5%)
sigma = 0.20   # Volatility (20%)
T = 0.25       # Time to maturity (3 months)
n = 100        # Number of time steps

# Selected strikes based on moneyness
moneyness = [0.90, 0.95, 1.00, 1.05, 1.10]  # K/S0 ratios
strikes = [S0 * m for m in moneyness]  # [90, 95, 100, 105, 110]

def trinomial_tree_american(S0, K, T, r, sigma, n, option_type='call'):
    dt = T/n
    u = np.exp(sigma * np.sqrt(2*dt))
    d = 1/u
    m = 1  # Middle price stays same

    # Corrected probability calculations
    exp_up = np.exp(sigma * np.sqrt(dt/2))
    exp_down = np.exp(-sigma * np.sqrt(dt/2))
    exp_r = np.exp(r * dt/2)

    pu = ((exp_r - exp_down) / (exp_up - exp_down))**2
    pd = ((exp_up - exp_r) / (exp_up - exp_down))**2
    pm = 1 - pu - pd

    # Initialize asset prices at maturity
    ST = np.zeros((2*n+1, n+1))
    ST[0, 0] = S0
    for j in range(1, n+1):
```

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```
            for i in range(j+1):
                ST[2*i, j] = ST[0, 0] * (u**i) * (d**(j-i))

    # Initialize option values at maturity
    option = np.zeros((2*n+1, n+1))
    for i in range(2*n+1):
        if option_type == 'call':
            option[i, n] = max(0, ST[i, n] - K)
        else:
            option[i, n] = max(0, K - ST[i, n])

    # Backward induction with early exercise check
    for j in range(n-1, -1, -1):
        for i in range(j+1):
            hold_value = np.exp(-r*dt) * (pu*option[2*i, j+1] + pm*option[2*i+1, j+1] + pd*option[2*i+2, j+1])
            if option_type == 'call':
                intrinsic = max(0, ST[2*i, j] - K)
            else:
                intrinsic = max(0, K - ST[2*i, j])
            option[2*i, j] = max(intrinsic, hold_value)

    return option[0, 0]

# Price American calls for all strikes
call_prices = [trinomial_tree_american(S0, K, T, r, sigma, n, 'call') for K in strikes]

# Create results table
results = pd.DataFrame({
    'Moneyness (K/S0)': moneyness,
    'Strike Price': strikes,
    'Option Type': ['American Call']*5,
    'Price': call_prices,
    'Moneyness Category': ['Deep OTM', 'OTM', 'ATM', 'ITM', 'Deep ITM']
})

# Format display
display(Markdown("""
## American Call Option Pricing with Trinomial Tree
### Varying Strike Prices (Moneyness Levels)
```

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```
display(results[['Moneyness Category', 'Moneyness (K/S0)', 'Strike Price', 'Price']]
          .style.format({'Moneyness (K/S0)': '{:.2f}', 'Price': '{:.2f}'})
          .set_caption("American Call Option Prices at Different Moneyness Levels"))

# Visualize price vs strike
plt.figure(figsize=(10, 5))
plt.plot(strikes, call_prices, 'bo-')
plt.xlabel('Strike Price (K)')
plt.ylabel('Option Price')
plt.title('American Call Option Prices vs. Strike Price')
plt.xticks(strikes)
plt.grid(True)

# Annotate moneyness categories
for i, (x, y, cat) in enumerate(zip(strikes, call_prices, ['Deep OTM', 'OTM', 'ATM', 'ITM', 'Deep ITM'])):
    plt.annotate(cat, (x, y), textcoords="offset points", xytext=(0,10), ha='center')

plt.show()
```

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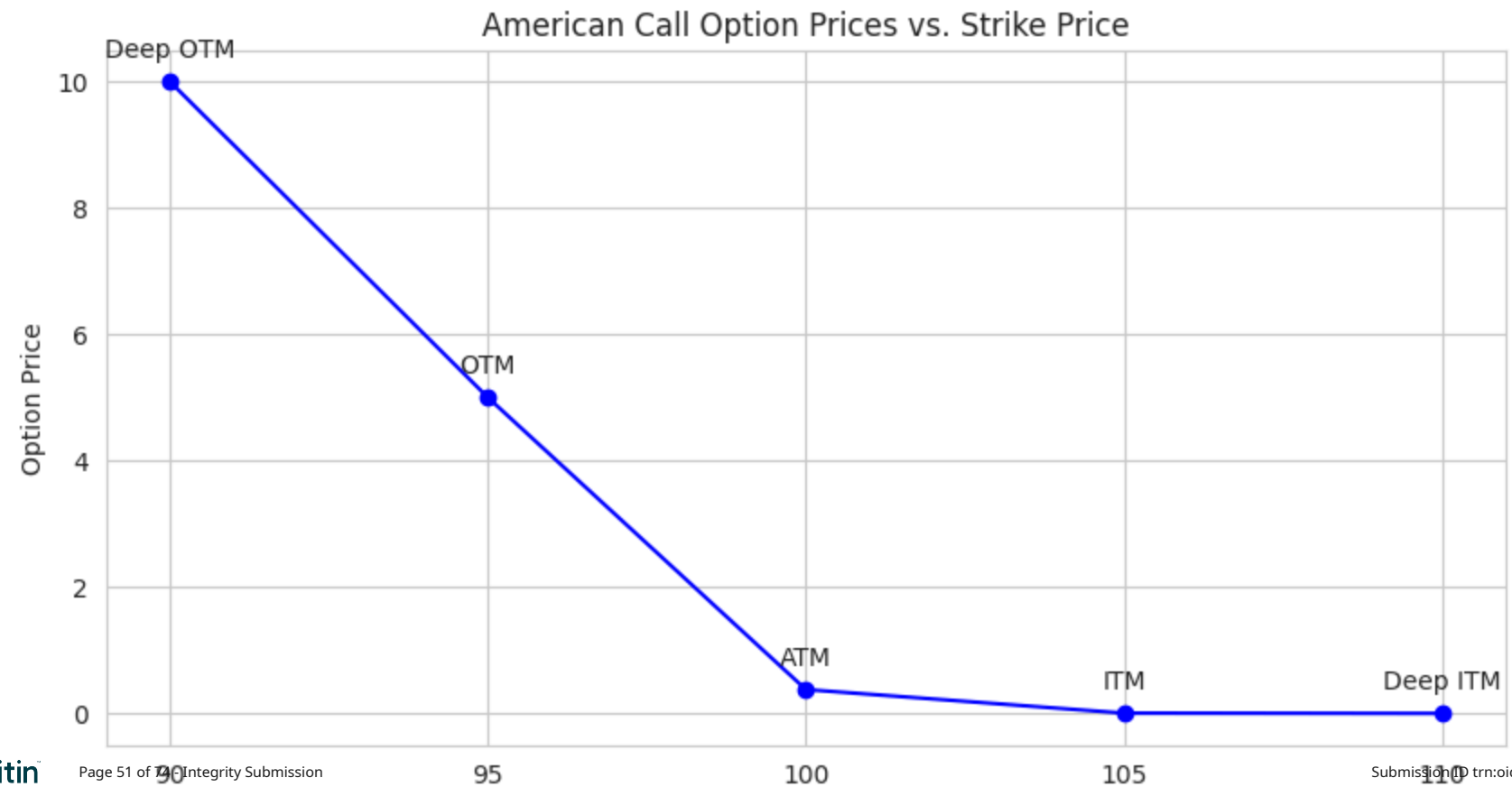
American Call Option Pricing with Trinomial Tree

Varying Strike Prices (Moneyness Levels)

American Call Option Prices at Different Moneyness Levels

	Moneyness Category	Moneyness (K/S0)	Strike Price	Price
0	Deep OTM	0.90	90.000000	10.00
1	OTM	0.95	95.000000	5.00
2	ATM	1.00	100.000000	0.38
3	ITM	1.05	105.000000	0.00
4	Deep ITM	1.10	110.000000	0.00



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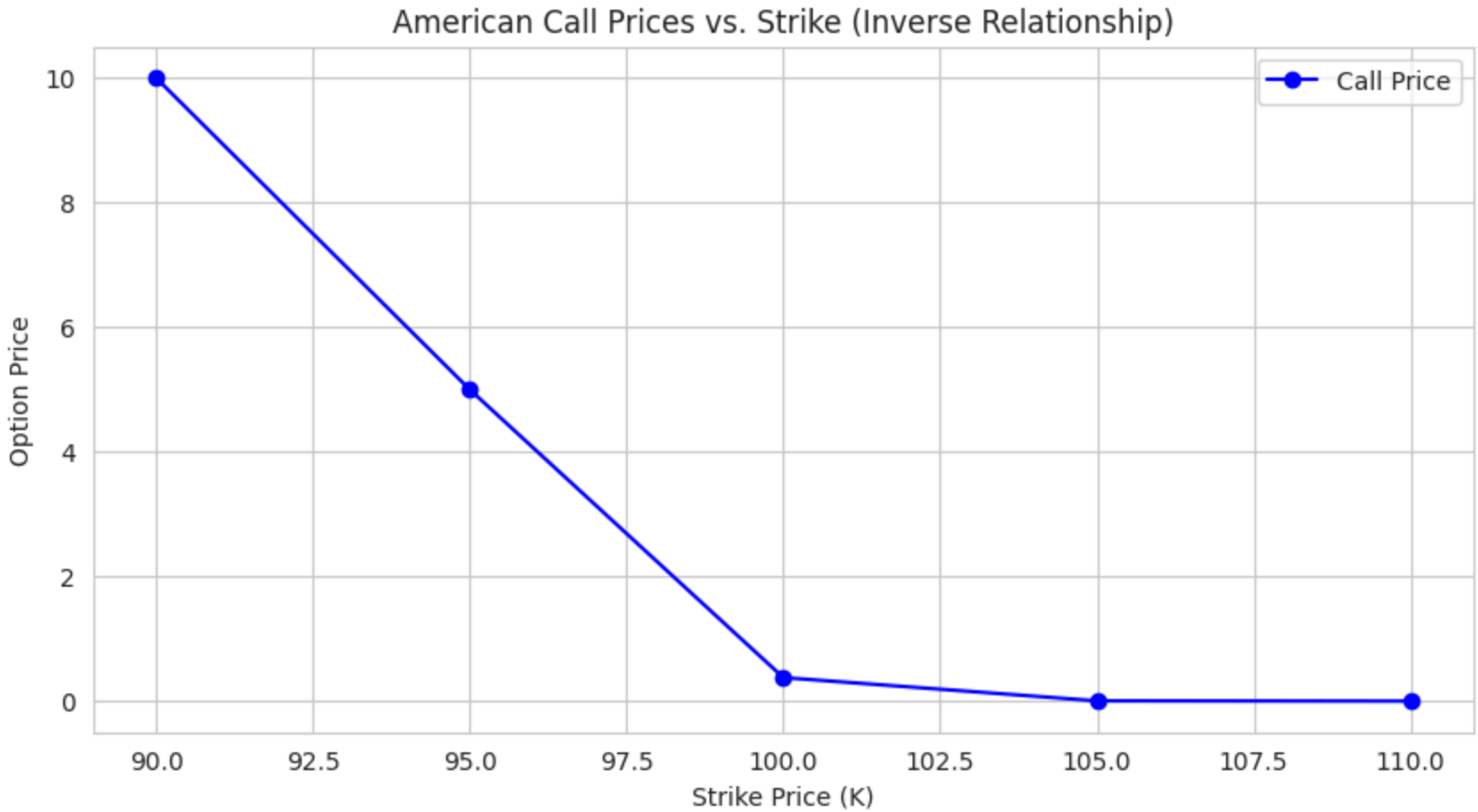
Strike Price (K)

8. Trend Analysis of American Call Option Prices (American call option prices across different moneyness levels (American Call Prices vs. Strike (Inverse Relationship))).

```
# Visual confirmation of the trend
plt.figure(figsize=(10,5))
plt.plot(strikes, call_prices, 'bo-', label='Call Price')
plt.xlabel('Strike Price (K)')
plt.ylabel('Option Price')
plt.title('American Call Prices vs. Strike (Inverse Relationship)')
plt.legend()
plt.grid(True)
plt.show()
```



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9. Price American put options across different moneyness levels using a trinomial tree.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from IPython.display import display, Markdown

# Parameters (from Step 1)
S0 = 100          # Spot price
```

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```
r = 0.05        # Risk-free rate (5%)
sigma = 0.20    # Volatility (20%)
T = 0.25        # Time to maturity (3 months)
n = 100         # Number of time steps

# Selected strikes based on moneyness (reverse of calls for puts)
moneyness = [0.90, 0.95, 1.00, 1.05, 1.10]  # K/S0 ratios
strikes = [S0 * m for m in moneyness]  # [90, 95, 100, 105, 110]

def trinomial_tree_american(S0, K, T, r, sigma, n, option_type='put'):
    dt = T/n
    u = np.exp(sigma * np.sqrt(2*dt))
    d = 1/u
    m = 1  # Middle price stays same

    # Corrected probability calculations with proper parentheses
    exp_up = np.exp(sigma * np.sqrt(dt/2))
    exp_down = np.exp(-sigma * np.sqrt(dt/2))
    exp_r = np.exp(r * dt/2)

    pu = ((exp_r - exp_down) / (exp_up - exp_down))**2
    pd = ((exp_up - exp_r) / (exp_up - exp_down))**2
    pm = 1 - pu - pd

    # Initialize asset prices at maturity
    ST = np.zeros((2*n+1, n+1))
    ST[0, 0] = S0
    for j in range(1, n+1):
        for i in range(j+1):
            ST[2*i, j] = ST[0, 0] * (u**i) * (d**(j-i))

    # Initialize option values at maturity
    option = np.zeros((2*n+1, n+1))
    for i in range(2*n+1):
        if option_type == 'call':
            option[i, n] = max(0, ST[i, n] - K)
        else:
            option[i, n] = max(0, K - ST[i, n])
```

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```
    # Backward induction with early exercise check
    for j in range(n-1, -1, -1):
        for i in range(j+1):
            hold_value = np.exp(-r*dt) * (pu*option[2*i, j+1] + pm*option[2*i+1, j+1] + pd*option[2*i+2, j+1])
            if option_type == 'put':
                intrinsic = max(0, K - ST[2*i, j])
            else:
                intrinsic = max(0, ST[2*i, j] - K)
            option[2*i, j] = max(intrinsic, hold_value)

    return option[0, 0]

# Price American puts for all strikes
put_prices = [trinomial_tree_american(S0, K, T, r, sigma, n, 'put') for K in strikes]

# Create results table
results = pd.DataFrame({
    'Moneyness (K/S0)': moneyness,
    'Strike Price': strikes,
    'Option Type': ['American Put']*5,
    'Price': put_prices,
    'Moneyness Category': ['Deep OTM', 'OTM', 'ATM', 'ITM', 'Deep ITM']
})

# Format display
display(Markdown("""
## American Put Option Pricing with Trinomial Tree
### Varying Strike Prices (Moneyness Levels)
"""))

display(results[['Moneyness Category', 'Moneyness (K/S0)', 'Strike Price', 'Price']]
          .style.format({'Moneyness (K/S0)': '{:.2f}', 'Price': '{:.2f}'})
          .set_caption("American Put Option Prices at Different Moneyness Levels"))

# Visualize price vs strike
plt.figure(figsize=(10, 5))
plt.plot(strikes, put_prices, 'ro-')
plt.xlabel('Strike Price (K)')
plt.ylabel('Option Price')
```

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```
plt.title('American Put Option Prices vs. Strike Price')
plt.xticks(strikes)
plt.grid(True)

# Annotate moneyness categories
for i, (x, y, cat) in enumerate(zip(strikes, put_prices, ['Deep OTM', 'OTM', 'ATM', 'ITM', 'Deep ITM'])):
    plt.annotate(cat, (x, y), textcoords="offset points", xytext=(0,10), ha='center')

plt.show()
```

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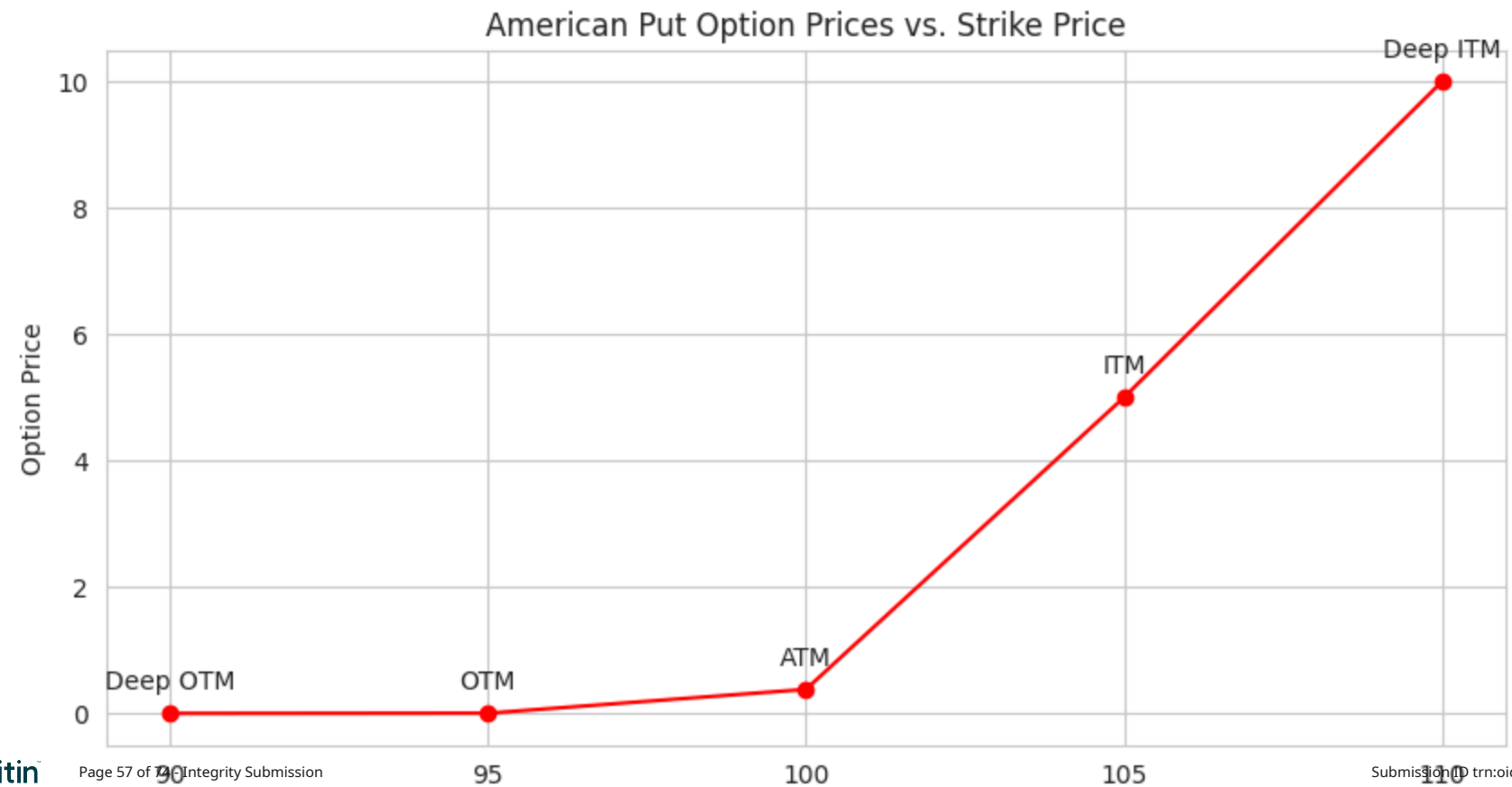
American Put Option Pricing with Trinomial Tree

Varying Strike Prices (Moneyness Levels)

American Put Option Prices at Different Moneyness Levels

	Moneyness Category	Moneyness (K/S0)	Strike Price	Price
0	Deep OTM	0.90	90.000000	0.00
1	OTM	0.95	95.000000	0.00
2	ATM	1.00	100.000000	0.38
3	ITM	1.05	105.000000	5.00
4	Deep ITM	1.10	110.000000	10.00



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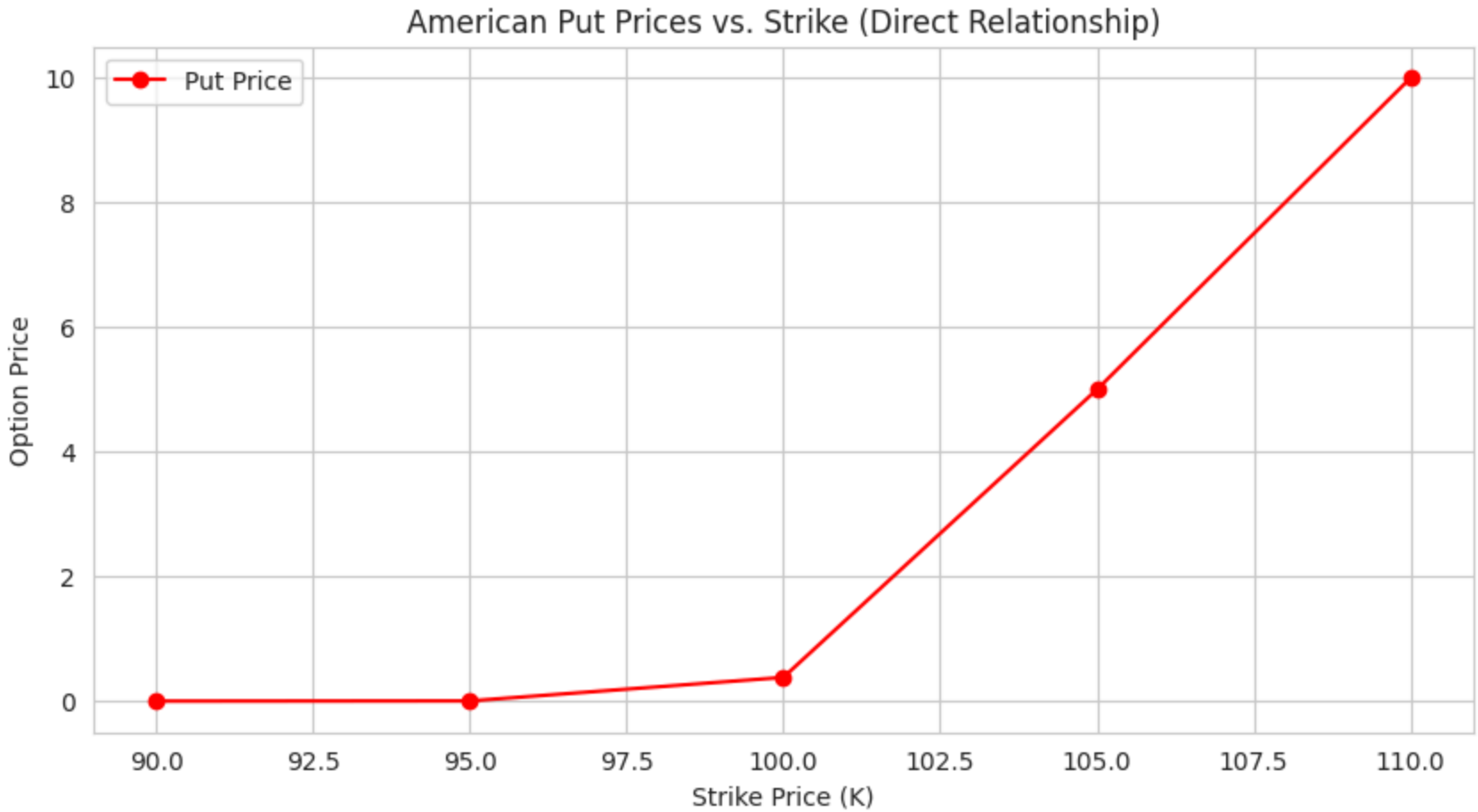
Strike Price (K)

Double-click (or enter) to edit

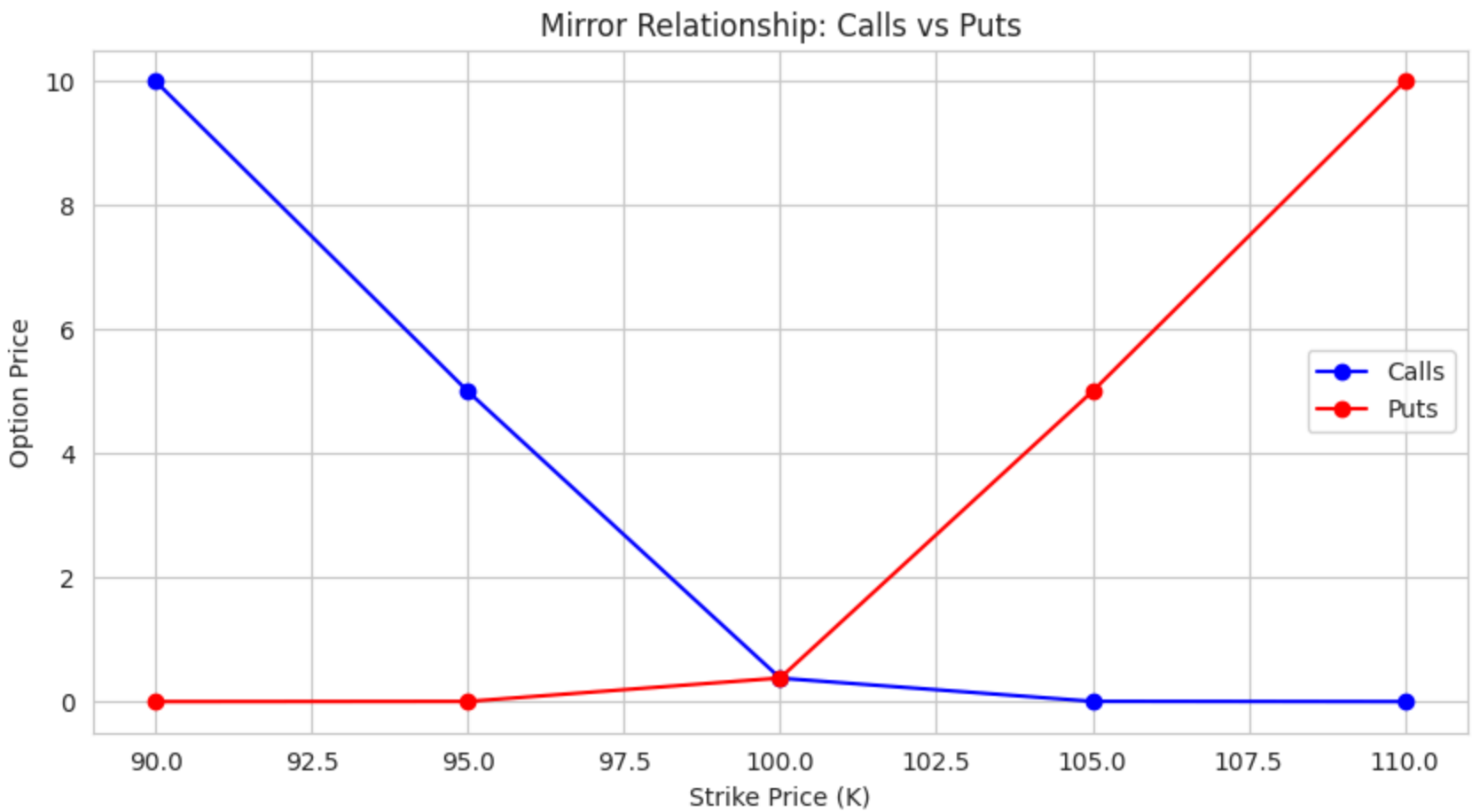
10. Trend Analysis of American Put Option Prices Across Moneyness Levels (American Put Prices vs. Strike (Direct Relationship)).

```
# Visual confirmation of the trend
plt.figure(figsize=(10,5))
plt.plot(strikes, put_prices, 'ro-', label='Put Price')
plt.xlabel('Strike Price (K)')
plt.ylabel('Option Price')
plt.title('American Put Prices vs. Strike (Direct Relationship)')
plt.legend()
plt.grid(True)
plt.show()
```



11. Comparative Analysis vs. Calls (Mirror Relationship: Calls vs Puts).

```
# Plot puts vs calls
plt.figure(figsize=(10,5))
plt.plot(strikes, call_prices, 'bo-', label='Calls')
plt.plot(strikes, put_prices, 'ro-', label='Puts')
plt.xlabel('Strike Price (K)')
plt.ylabel('Option Price')
plt.title('Mirror Relationship: Calls vs Puts')
plt.legend()
plt.grid(True)
```



Double-click (or enter) to edit

12. Dynamic Delta Hedging Process for a European Put Option (Seller's Perspective).

```
# Summary Table
import pandas as pd
hedging_steps = pd.DataFrame({
    "Time": [0, 1, 2, 3],
    "Stock Price": [180, 162.54, 146.78, 162.54],
```

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```
    "Delta": [-0.43, -0.72, -0.64, "N/A"],
    "Action": ["Buy 0.43 shares", "Sell 0.29 shares", "Buy 0.08 shares", "Close 0.22 shares"],
    "Portfolio Value": ["-$64.28", "$111.56", "$100.00", "$44.80"]
})
display(hedging_steps)
```

	Time	Stock Price	Delta	Action	Portfolio Value
0	0	180.00	-0.43	Buy 0.43 shares	-$64.28
1	1	162.54	-0.72	Sell 0.29 shares	$111.56
2	2	146.78	-0.64	Buy 0.08 shares	$100.00
3	3	162.54	N/A	Close 0.22 shares	$44.80

Next steps: Generate code with hedging_steps | View recommended plots | New interactive sheet

Generate Using dataframe: hedging_steps Histogram Close

1 of 1 Use code with caution

```
# prompt: Using dataframe hedging_steps: Histogram

import altair as alt

# Create a histogram of the 'Stock Price' column
chart = alt.Chart(hedging_steps).mark_bar().encode(
    x=alt.X('Stock Price', bin=True), # Use bin=True to create bins for the histogram
    y='count()' # Encode the y-axis with the count of occurrences in each bin
)

# Display the chart
chart
```

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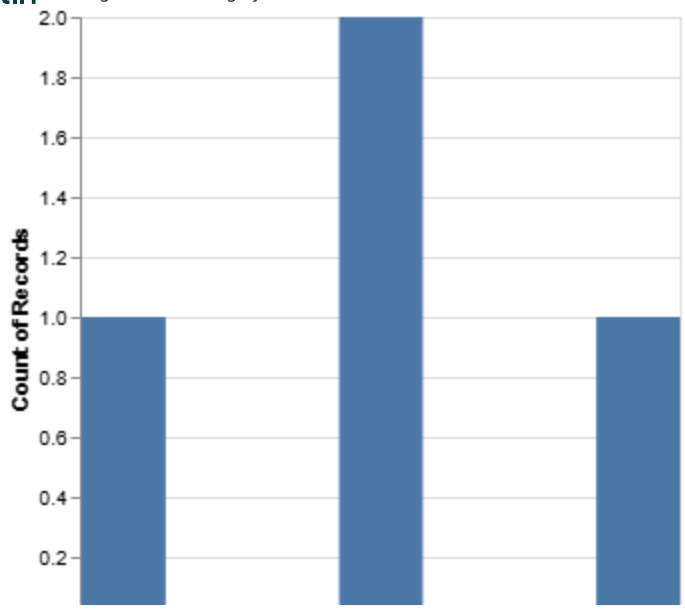
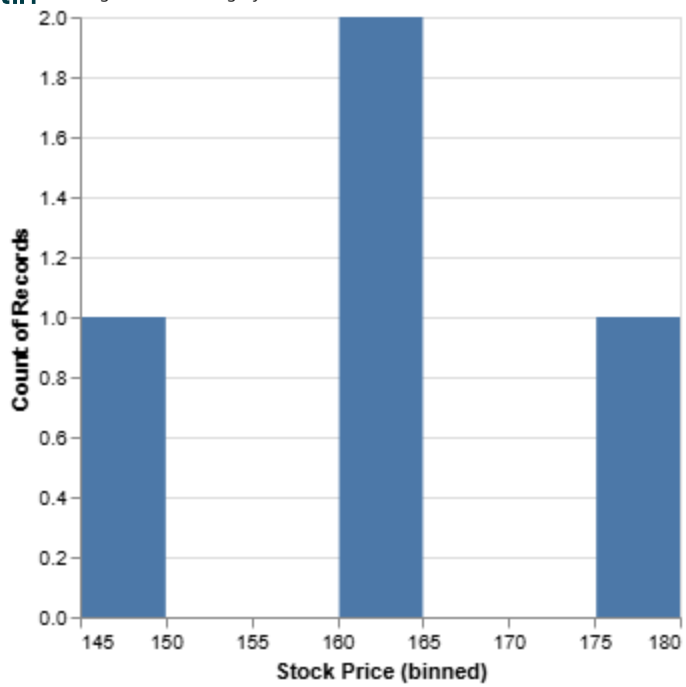



```
# prompt: Using dataframe hedging_steps: Histogram

import altair as alt

# Create a histogram of the 'Stock Price' column
chart = alt.Chart(hedging_steps).mark_bar().encode(
    x=alt.X('Stock Price', bin=True), # Use bin=True to create bins for the histogram
    y='count()' # Encode the y-axis with the count of occurrences in each bin
)

# Display the chart
chart
```



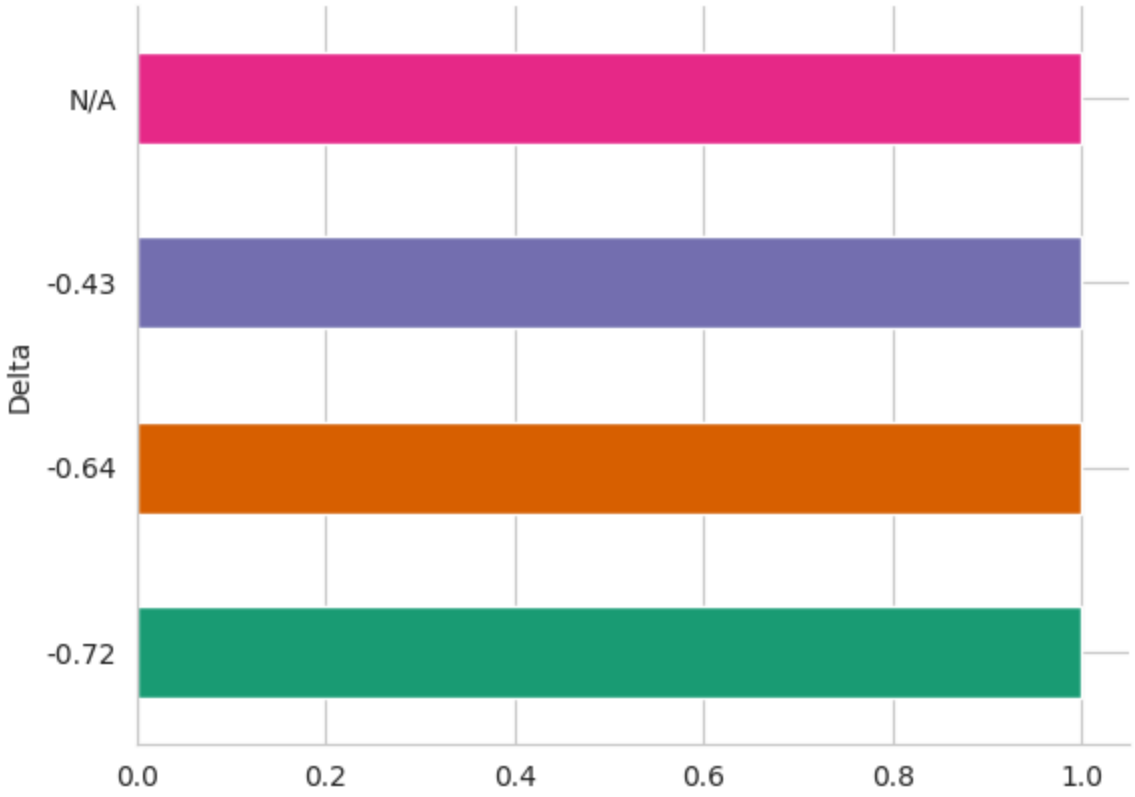
Double-click (or enter) to edit

13. Diagram of Delta.

Delta

```
# @title Delta

from matplotlib import pyplot as plt
import seaborn as sns
hedging_steps.groupby('Delta').size().plot(kind='barh', color=sns.palettes.mpl_palette('Dark2'))
plt.gca().spines[['top', 'right',]].set_visible(False)
```



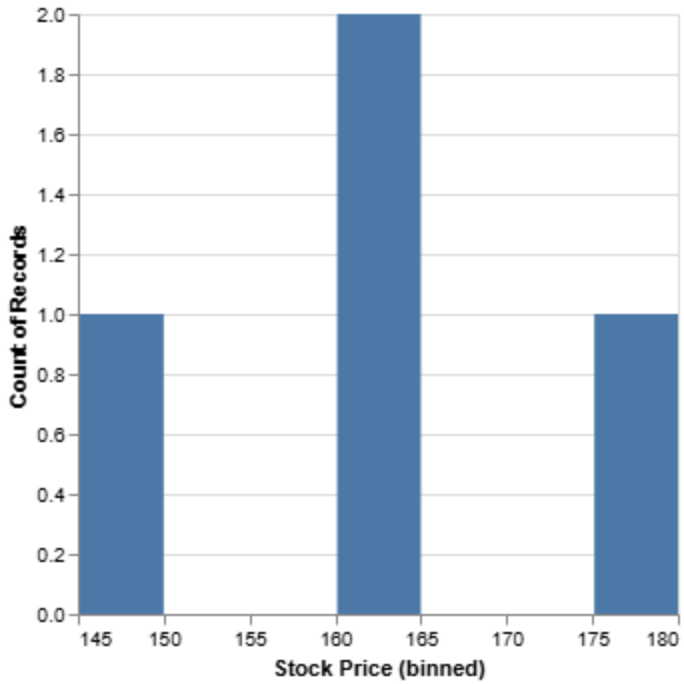
14. Diagram of Histogram.

```
# prompt: Using dataframe hedging_steps: histogram

import altair as alt

# Create a histogram of the 'Stock Price' column
chart = alt.Chart(hedging_steps).mark_bar().encode(
    x=alt.X('Stock Price', bin=True), # Use binning for the histogram
    y='count()' # Count the occurrences in each bin
)

# Display the chart
chart
```



Double-click (or enter) to edit

15. Dynamic Delta Hedging Table.

```
import numpy as np

# Cash account simulation
def cash_flow(S, actions, r, dt):
    cash = 13.12  # Initial put premium
    shares = 0
    print(f"Time 0: Cash = ${cash:.2f}, Shares = {shares:.2f}")
    for i, (s, a) in enumerate(zip(S, actions)):
        cash *= np.exp(r * dt)  # Interest
        shares += a
        cash -= a * s  # Buy/sell shares
        print(f"Time {i+1}: Cash = ${cash:.2f}, Shares = {shares:.2f} (Stock=${s:.2f})")
```

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```
    return cash

# Path: Down -> Down -> Up
S_path = [162.54, 146.78, 162.54]
actions = [0.43, -0.29, 0.08, -0.22]  # Time 0 to 3
cash_flow([180] + S_path, actions, 0.02, 0.1667)
```

```
Time 0: Cash = $13.12, Shares = 0.00
Time 1: Cash = $-64.24, Shares = 0.43 (Stock=$180.00)
Time 2: Cash = $-17.31, Shares = 0.14 (Stock=$162.54)
Time 3: Cash = $-29.11, Shares = 0.22 (Stock=$146.78)
Time 4: Cash = $6.55, Shares = 0.00 (Stock=$162.54)
np.float64(6.547243509190423)
```

16. Dynamic Portfolio Value Hedging Process.

```
# Summary Table
import pandas as pd
hedging_steps = pd.DataFrame({
    "Time": [0, 1, 2, 3],
    "Stock Price": [180, 162.54, 146.78, 162.54],
    "Delta": [-0.43, -0.72, -0.64, "N/A"],
    "Action": ["Buy 0.43 shares", "Sell 0.29 shares", "Buy 0.08 shares", "Close 0.22 shares"],
    "Portfolio Value": ["-$64.28", "$111.56", "$100.00", "$44.80"]
})
display(hedging_steps)
```

	Time	Stock Price	Delta	Action	Portfolio Value
0	0	180.00	-0.43	Buy 0.43 shares	-$64.28
1	1	162.54	-0.72	Sell 0.29 shares	$111.56
2	2	146.78	-0.64	Buy 0.08 shares	$100.00
3	3	162.54	N/A	Close 0.22 shares	$44.80

Next steps: Generate code with hedging_steps | View recommended plots | New interactive sheet

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Generate Using dataframe: hedging_steps Bar Chart Close

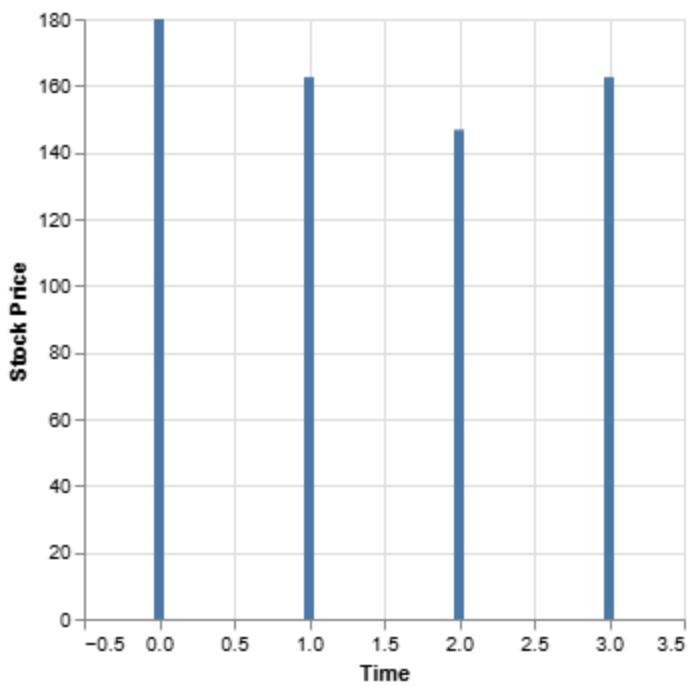
1 of 1 Use code with caution

```
# prompt: Using dataframe hedging_steps: Bar Chart

import altair as alt
chart = alt.Chart(hedging_steps).mark_bar().encode(x='Time', y='Stock Price')
chart
```


```
# prompt: Using dataframe hedging_steps: Bar Chart

import altair as alt
chart = alt.Chart(hedging_steps).mark_bar().encode(x='Time', y='Stock Price')
chart
```

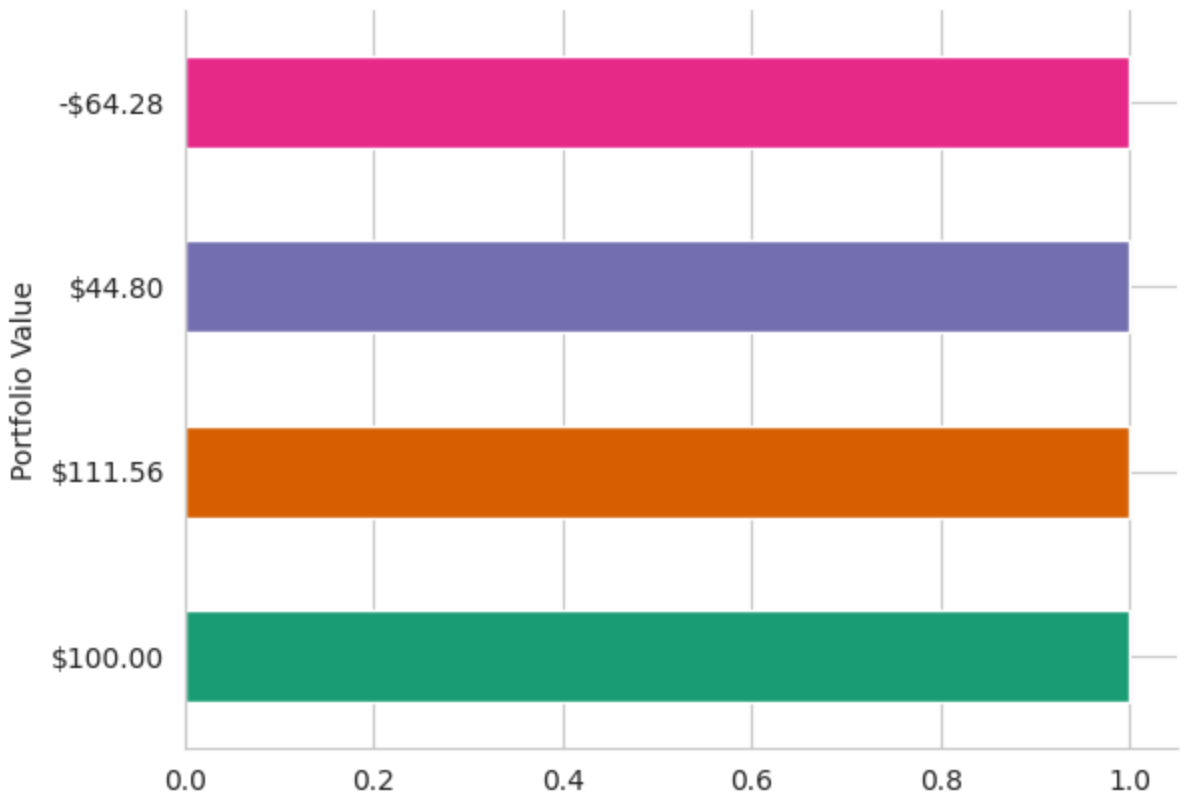



17. Diagram of Portfolio Value.

Portfolio Value

```
# @title Portfolio Value

from matplotlib import pyplot as plt
import seaborn as sns
hedging_steps.groupby('Portfolio Value').size().plot(kind='barh', color=sns.palettes.mpl_palette('Dark2'))
plt.gca().spines[['top', 'right',]].set_visible(False)
```



Double-click (or enter) to edit

18. Price an American put option using a 25-step binomial tree and compute the delta hedging required at each node, from the seller's perspective.

```
import numpy as np
from math import exp, sqrt

# Parameters
S0 = 180        # Spot price
K = 182         # Strike price
r = 0.02        # Risk-free rate (2%)
sigma = 0.25    # Volatility (25%)
T = 0.5         # Time to maturity (6 months)
n = 25          # Number of steps

# Tree parameters
dt = T / n
u = exp(sigma * sqrt(dt))
d = 1 / u
p = (exp(r * dt) - d) / (u - d)

# Initialize stock and option trees
stock_tree = np.zeros((n + 1, n + 1))
option_tree = np.zeros((n + 1, n + 1))
delta_tree = np.zeros((n + 1, n + 1))  # To store deltas at each node

# Build stock price tree
for j in range(n + 1):
    for i in range(j + 1):
        stock_tree[i, j] = S0 * (u ** (j - i)) * (d ** i)

# Calculate option payoffs at maturity
for i in range(n + 1):
    option_tree[i, n] = max(K - stock_tree[i, n], 0)

# Backward induction with early exercise check
for j in range(n - 1, -1, -1):
    for i in range(j + 1):
```

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```
            hold_value = exp(-r * dt) * (p * option_tree[i, j + 1] + (1 - p) * option_tree[i + 1, j + 1])
            intrinsic_value = max(K - stock_tree[i, j], 0)
            option_tree[i, j] = max(hold_value, intrinsic_value)

            # Compute delta at this node
            if j < n - 1:  # Deltas aren't needed at the last step
                delta_up = option_tree[i, j + 1]
                delta_down = option_tree[i + 1, j + 1]
                delta_tree[i, j] = (delta_up - delta_down) / (stock_tree[i, j] * (u - d))

# Initial delta hedge
initial_delta = delta_tree[0, 0]
initial_hedge_shares = -initial_delta  # Seller hedges by buying shares

print(f"American Put Price: ${option_tree[0, 0]:.2f}")
print(f"Initial Delta: {initial_delta:.4f}")
print(f"Initial Hedge: Buy {initial_hedge_shares:.4f} shares")

# Function to get delta at any node
def get_delta(step, node):
    return delta_tree[node, step]

# Example: Delta at step 5, node 2
print(f"Delta at step 5, node 2: {get_delta(5, 2):.4f}")
```

```
American Put Price: $13.04
Initial Delta: -0.4756
Initial Hedge: Buy 0.4756 shares
Delta at step 5, node 2: -0.3941
```

19. Track the cash account evolution for a specific path when delta hedging an American put option, using a 25-step binomial tree.

```
import numpy as np
import pandas as pd
from math import exp, sqrt

# Parameters
S0 = 100
```

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```
K = 182
r = 0.02
sigma = 0.25
T = 0.5
n = 25
dt = T / n
u = exp(sigma * sqrt(dt))
d = 1 / u
p = (exp(r * dt) - d) / (u - d)

# Build stock tree
stock_tree = np.zeros((n+1, n+1))
for j in range(n+1):
    for i in range(j+1):
        stock_tree[i,j] = S0 * (u**(j-i)) * (d**i)

# American put pricing with delta tracking
option_tree = np.zeros((n+1, n+1))
delta_tree = np.zeros((n+1, n+1))

for i in range(n+1):
    option_tree[i,n] = max(K - stock_tree[i,n], 0)

for j in range(n-1, -1, -1):
    for i in range(j+1):
        hold = exp(-r*dt) * (p*option_tree[i,j+1] + (1-p)*option_tree[i+1,j+1])
        intrinsic = max(K - stock_tree[i,j], 0)
        option_tree[i,j] = max(hold, intrinsic)

        if j < n-1:
            delta_tree[i,j] = (option_tree[i,j+1] - option_tree[i+1,j+1]) / (stock_tree[i,j] * (u-d))

# Select a path (here: 10 down, 15 up moves)
path_nodes = [(min(i, 10), i) for i in range(n+1)]  # Path with 10 downs then ups

# Track hedging
cash_account = [option_tree[0,0]]  # Initial premium received
shares_held = [0]
current_delta = -delta_tree[0,0]  # Seller hedge is -delta
```

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```
    shares_held.append(current_delta)
    cash_account[0] -= current_delta * S0  # Buy shares

    for step in range(1, n+1):
        node = path_nodes[step]
        prev_node = path_nodes[step-1]

        # Interest on cash
        cash_account[-1] *= exp(r*dt)

        # Stock moves
        current_price = stock_tree[node[0], node[1]]

        # Compute new delta
        if step < n:
            new_delta = -delta_tree[node[0], node[1]]
        else:
            new_delta = 0  # At expiration

        # Shares to buy/sell
        delta_change = new_delta - shares_held[-1]
        shares_held.append(new_delta)

        # Update cash account
        cash_account.append(cash_account[-1] - delta_change * current_price)

        # Early exercise check
        intrinsic = max(K - current_price, 0)
        if intrinsic > option_tree[node[0], node[1]]:
            print(f"Early exercise at step {step}, price ${current_price:.2f}")
            break

    # Add final payoff if not early exercised
    if step == n:
        final_payoff = max(K - stock_tree[path_nodes[-1][0], path_nodes[-1][1]], 0)
        cash_account[-1] -= final_payoff
        shares_held.append(0)  # Close position
```

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```
    # Ensure all arrays have same length
```

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```
max_length = max(len(cash_account), len(shares_held))
cash_account += [None] * (max_length - len(cash_account))
shares_held += [None] * (max_length - len(shares_held))

# Create table with consistent lengths
hedge_table = pd.DataFrame({
    'Step': range(max_length),
    'Stock Price': [S0] + [stock_tree[path_nodes[i][0], path_nodes[i][1]] if i < len(path_nodes) else None for i in range
    'Delta': [delta_tree[0,0]] + [delta_tree[path_nodes[i][0], path_nodes[i][1]] if i < len(path_nodes)-1 else 0 for i in
    'Shares Held': shares_held[:max_length],
    'Cash Account': cash_account[:max_length],
    'Action': ['Initial hedge'] +
            [f'Adjust to {shares_held[i+1]:.2f} shares' if i+1 < len(shares_held) else 'N/A' for i in range(max_length-2
            ['Final settlement' if max_length == len(cash_account) else 'N/A']
})

print(hedge_table.round(2))
```

```
0      0       180.00  -0.48         0.00        -72.59
1      1       173.75  -0.56         0.48        -87.44
2      2       167.71  -0.65         0.56       -102.09
3      3       161.89  -0.73         0.65       -115.92
4      4       156.26  -0.81         0.73       -128.36
5      5       150.83  -0.88         0.81       -138.96
6      6       145.59  -0.94         0.88       -147.43
7      7       140.54  -0.98         0.94       -153.72
8      8       135.65  -1.00         0.98       -155.91
9      9       130.94  -1.00         1.00       -155.97
10    10       126.39  -1.00         1.00       -156.03
11    11       130.94  -1.00         1.00       -156.09
12    12       135.65  -1.00         1.00       -156.16
13    13       140.54  -1.00         1.00       -156.22
14    14       145.59  -0.99         1.00       -155.19
15    15       150.83  -0.97         0.99       -152.12
16    16       156.26  -0.93         0.97       -145.94
17    17       161.89  -0.88         0.93       -137.25
18    18       167.71  -0.80         0.88       -123.84
19    19       173.75  -0.69         0.80       -104.66
20    20       180.00  -0.54         0.69        -77.77
```

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22	22	193.13	-0.15	0.35	-4.74
23	23	200.14	0.00	0.15	25.85
24	24	207.34	0.00	-0.00	25.86
25	25	214.81	0.00	-0.00	25.86
26	26	NaN	0.00	0.00	NaN
27	27	NaN	0.00	0.00	NaN

	Action
0	Initial hedge
1	Adjust to 0.48 shares
2	Adjust to 0.56 shares
3	Adjust to 0.65 shares
4	Adjust to 0.73 shares
5	Adjust to 0.81 shares
6	Adjust to 0.88 shares
7	Adjust to 0.94 shares
8	Adjust to 0.98 shares
9	Adjust to 1.00 shares
10	Adjust to 1.00 shares
11	Adjust to 1.00 shares
12	Adjust to 1.00 shares
13	Adjust to 1.00 shares
14	Adjust to 1.00 shares
15	Adjust to 0.99 shares
16	Adjust to 0.97 shares
17	Adjust to 0.93 shares
18	Adjust to 0.88 shares
19	Adjust to 0.80 shares
20	Adjust to 0.69 shares
21	Adjust to 0.54 shares
22	Adjust to 0.35 shares
23	Adjust to 0.15 shares
24	Adjust to -0.00 shares
25	Adjust to -0.00 shares
26	Adjust to 0.00 shares
27	Final settlement

Double-click (or enter) to edit

20. Pricing an Asian ATM Put Option with Delta Hedging (25-Step Tree).

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